

# Automatic Pronunciation Assessment - A Review

Yassine El Kheir, Ahmed Ali and Shammur Absar Chowdhury

Qatar Computing Research Institute, HBKU, Qatar

shchowdhury@hbku.edu.qa

## Abstract

Pronunciation assessment and its application in computer-aided pronunciation training (CAPT) have seen impressive progress in recent years. With the rapid growth in language processing and deep learning over the past few years, there is a need for an updated review. In this paper, we review methods employed in pronunciation assessment for both phonemic and prosodic. We categorize the main challenges observed in prominent research trends, and highlight existing limitations, and available resources. This is followed by a discussion of the remaining challenges and possible directions for future work.

## 1 Introduction

Computer-aided Pronunciation Training (CAPT) technologies are pivotal in promoting self-directed language learning, offering constant, and tailored feedback for secondary language learners. The rising demand for foreign language learning, with the tide of globalization, fuels the increment in the development of CAPT systems. This surge has led to extensive research and development efforts in the field (Neri et al., 2008; Kang et al., 2018; Rogerson-Revell, 2021). CAPT systems have two main usages: (i) pronunciation assessment, where the system is concerned with the errors in the speech segment; (ii) pronunciation teaching, where the system is concerned with correcting and guiding the learner to fix mistakes in their pronunciation.

This paper addresses the former – focusing on pronunciation assessment, which aims to automatically score non-native speech-segment and give meaningful feedback. To build such a robust pronunciation assessment system, the following design aspects should be addressed.

**Modelling** Mispronunciation detection and diagnosis (MDD), in many cases, are more challenging to model compared to the vanilla automatic speech recognition (ASR) system, which converts speech

into text regardless of pronunciation mistakes. Robust ASR should perform well with all variation including dialects and non-native speakers. However, MDD should mark phonetic variations from the learner, which may sometimes be subtle differences (Li et al., 2016a).

**Training Resources** Recent success in deep learning methods emphasized the need for in-domain training data. Language learners can be divided into two groups: adult secondary (L2) language learners and children language learners – the former depends on whether to build a system that is native language dependant (L1). At the same time, the latter identifies the need for children’s voice, which is a challenging corpus to build (Council III et al., 2019; Venkatasubramaniam et al., 2023), even the accuracy for ASR for children is still behind compared to adult ASR (Liao et al., 2015). The scarcity and imbalanced distribution of negative mispronunciation classes pose a significant challenge in training data.

**Evaluation** There is no clear definition of right or wrong in pronunciation, instead an entire scale from unintelligible to native-sounding speech (Witt, 2012). Given that error in pronunciation is difficult to quantify, it can be split into (a) *Objective evaluations* – (i): phonetic or segmental; (ii): prosodic or supra-segmental; and (iii) place or articulation, manner of speech or sub-segmental; (b) *Subjective evaluations*; in many cases measured through listening tasks followed by human judgment, and can be split into three main classes: (i) intelligibility; (ii) comprehensibility and (iii) accentedness (or linguistic native-likeness). See Figure 1 for common pronunciation assessment factors.

Several studies have summarized advances in pronunciation error detection (Eskenazi, 1999, 2009; Witt, 2012; Li et al., 2016a; Chen and Li, 2016; Zhang et al., 2020; Caro Anzola and Mendoza Moreno, 2023). Eskenazi (1999) investigated

# 自动发音评估——综述

Yassine El Kheir、Ahmed Ali与Shammur Absar Chowdhury卡

塔尔计算研究中心，卡塔尔 HBKU shchowdhury@hbku.edu.qa

## 摘要

发音评估及其在计算机辅助发音训练 (CAPT) 中的应用近年来取得了显著进展。

随着近年来语言处理与深度学习领域的快速发展，亟需对相关研究进行一次更新后的综述。本文对用于音系学与韵律学发音评估的各种方法进行了系统性回顾。

我们对主流研究趋势中所面临的主要挑战进行了分类，并重点指出了当前存在的局限性及可用资源；随后，本文对尚待解决的挑战以及未来研究可能的发展方向进行了探讨。

## 1 引言

计算机辅助发音训练 (CAPT) 技术在促进自主语言学习方面发挥着关键作用，为中级语言学习者提供持续且个性化的反馈。随着全球化浪潮推动，对外语学习需求的日益增长促进了 CAPT 系统开发的加速发展。这一趋势引发了该领域的广泛研究与开发活动 (Neri 等, 2008; Kang 等, 2018; Rogerson Revell, 2021)。CAPT 系统主要有两种应用场景：

(i) 发音评估——系统针对语音片段中的发音错误进行分析；(ii) 发音教学——系统负责纠正并引导学习者修正其发音中的错误。

本文探讨前者——重点聚焦于发音评估，旨在对非母语语音片段进行自动评分并提供具有意义的反馈。为构建这样一个稳健的发音评估系统，需考虑以下设计方面。

**建模：**与无需考虑发音错误即可将语音转换为文本的常规自动语音识别 (ASR) 系统相比，误读检测与诊断 (MDD) 在建模上往往更具

挑战性。鲁棒的 ASR 系统应能良好应对所有语音变体，包括方言及非母语使用者的发音。然而，MDD 系统应能够识别学习者所呈现的语音变体，这些变体有时可能极为细微 (Li 等, 2016a)。

**训练资源**近期深度学习方法所取得的成功凸显了对领域内训练数据的需求。语言学习者可分为两类：成人第二语言 (L2) 学习者与儿童语言学习者——前者取决于是否构建一个依赖母语 (L1) 的系统；与此同时，后者凸显了对儿童语音数据集的需求——构建此类语料库具有挑战性 (Council III 等, 2019; Venkatasubramanian 等, 2023)，且与成人语音识别系统相比，针对儿童的语音识别准确率仍处于较低水平 (Liao 等, 2015)。负面错误发音类别的稀缺性及分布不均衡性给训练数据的获取带来了重大挑战。

**评估**语音发音方面并不存在明确的“正确”或“错误”之分，而是采用了一个从难以听懂到听起来像母语般的语音的完整量表 (Witt, 2012)。鉴于发音错误难以量化，可将其划分为：(a) 客观性评估——(i)：语音学或音段层面；(ii)：韵律学或超音段层面；以及 (iii)：发音部位或发音方式、说话方式或亚音段层面；(b) 主观性评估；在许多情况下，此类评估通过听觉任务后辅以人工判断来完成，可划分为三大类：(i) 可懂度；(ii) 可理解性与 (iii) 重音特征（或语言母语化程度）。常见发音评估因素请参阅图1。

多项研究综述了发音错误检测领域的最新进展 (Eskenazi, 1999、2009; Witt, 2012; Li 等, 2016a; Chen 与 Li, 2016; Zhang 等, 2020; Caro Anzola 与 Men Doza Moreno,

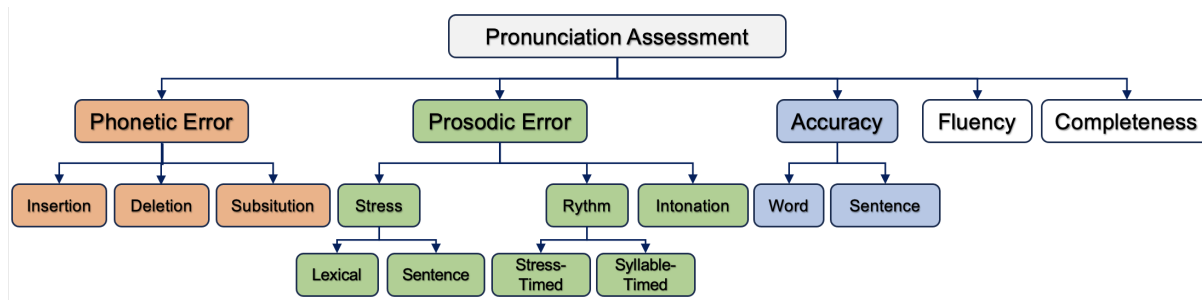


Figure 1: Types of Pronunciation Errors for Assessment

the potentials and limitations of ASR for L2 pronunciation assessment, showcasing its practical implementation using an interface developed at CMU. Furthermore, the study reports different automatic scoring techniques, emphasizing modalities of interaction, associated algorithms, and the challenges. Witt (2012) presented an overview of pronunciation error detection, encompassing various scoring methodologies and assessing commercial CAPT systems. Chen and Li (2016) provided a research summary, focusing on phoneme errors and prosodic error detection. More recently, Zhang et al. (2020) provided a summary of two automatic scoring approaches (a) ASR-based scoring to calculate confidence measures; and (b) acoustic phonetics scoring focusing on comparing or classifying phonetic segments using various acoustic features.

With large transformer-based pre-trained models gaining popularity, re-visiting the existing literature and presenting a comprehensive study of the field is timely. We provide an overview of techniques adapted for detecting mispronunciation in (a) segmental space, (b) assessing pronunciation with supra-segmental measures, along with (c) different data generation/augmentation approaches. Unlike previous overview studies, we also cover a handful of (d) qualitative studies bringing together the notion of intelligibility, comprehensiveness, and accentedness. We note the resources and evaluation measures available to the speech community and discuss the main challenges observed within prominent research trends, shedding light on existing limitations. Additionally, we also explore potential directions for future work.

## 2 Nuances of Pronunciation

Pronunciation can be defined as “the way in which a word or letter is said, or said correctly, or the way in which a language is spoken”.<sup>1</sup> Compared

<sup>1</sup><https://dictionary.cambridge.org/dictionary/english/pronunciation>, Accessed: 2023-06-21

to other language skills, learning pronunciation is difficult. Yet, for learners, mastering L2 pronunciation is most crucial for better communication. Historically, pronunciation errors (mispronunciations) are characterized by phonetic (segmental) errors and prosodic (supra-segmental) errors (Witt, 2012; Chen and Li, 2016), as represented in Figure 1. This characterization provides some clear distinctions for pronunciation assessment.

### 2.1 Pronunciation Errors

#### Phonetic Errors

Phonetic (segmental) errors involve the production of individual sounds, such as vowels, and consonants, and it includes three errors: insertion, deletion, and substitution. This can be attributed to several factors, including negative language transfer, incorrect letter-to-sound conversion, and misreading of text prompts (Meng et al., 2007b; Qian et al., 2010; Kartushina and Frauenfelder, 2014; Li et al., 2016a). For example, Arabic L1 speakers may find it difficult to differentiate between /p/ and /b/ as the phoneme /p/ is non-existent in Arabic, so verbs like /park/ and /bark/ might sound similar to Arabic L1 speakers. Similarly, in Spanish, there are no short vowels, so words like /eat/ and /it/ might sound similar to Spanish L1 speakers.

#### Prosodic Errors

Prosodic features encompass elements that influence the pronunciation of an entire word or sentence, including stress, rhythm, and intonation. Errors related to prosodic features involve the production of larger sound units. For intelligibility, prosodic features particularly play a significant role (Raux and Kawahara, 2002). This is especially true for tonal languages (Dahmen et al., 2023) where variation in the pitch can lead to words with different meanings. Prosodic errors are often language-dependent and categorized by: stress (lexical and sentence), rhythm, and intonation.

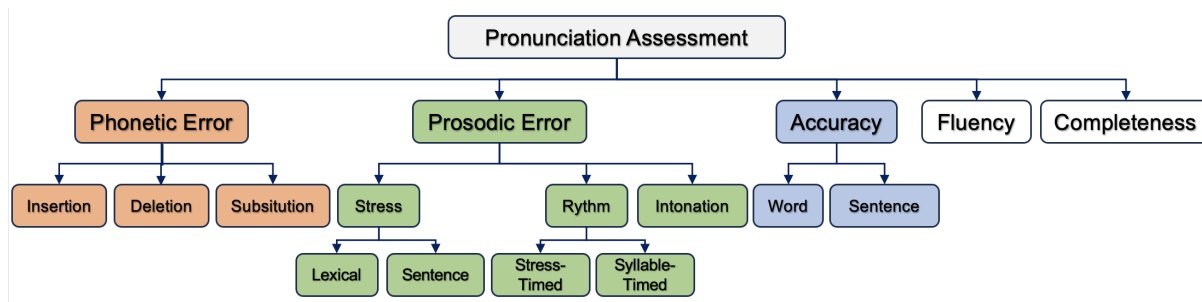


图 1：用于评估的发音错误类型

2023)。Eskenazi (1999) 探讨了自动语音识别 (ASR) 在第二语言发音评估中的应用潜力与局限性，并通过在卡内基梅隆大学开发的界面展示了其实际应用案例。此外，该研究还介绍了多种自动评分技术，重点阐述了交互模式、相关算法及所面临的挑战。Witt (2012) 对发音错误检测进行了综述，涵盖了多种评分方法并对商用 CAPT 系统进行了评估。Chen 与 Li (2016) 提供了研究报告摘要，聚焦于音素错误与韵律错误检测。近期，Zhang 等(2020) 对两种自动评分方法进行了概述：(a) 基于自动语音识别 (ASR) 的评分法，用于计算置信度度量；以及 (b) 声学语音学评分法，侧重于利用多种声学特征对语音片段进行比较或分类。

随着基于大型变压器的预训练模型日益普及，重新审视现有文献并对该领域开展全面研究已十分适时。本文概述了适用于检测发音错误的技术——包括 (a) 基于分段空间的检测方法、(b) 利用超分段度量进行发音评估，以及 (c) 多种数据生成与数据增强方法。与以往的综述性研究不同，本文还涵盖了若干项 (d) 定性研究，这些研究综合了可懂度、全面性与重音度等概念。我们列举了语音学界可利用的资源与评估指标，并探讨了在主流研究趋势中观察到的主要挑战，从而揭示了当前研究存在的局限性。此外，本文还探讨了未来研究可能的发展方向。

## 2 发音细微差别

发音可定义为“单词或字母的读音方式、正确读音方式，或语言的口语表达方式”。<sup>1</sup>与其

<sup>1</sup><https://dictionary.cambridge.org/dictionary/英语/发音>，访问日期：2023-06-21

他语言技能相比，学习发音较为困难。然而，对于学习者而言，掌握第二语言 (L2) 的发音对于实现更有效的沟通至关重要。从历史角度来看，发音错误（误读）通常表现为语音（段落性）错误与韵律（超段落性）错误 (Witt, 2012; Chen 与 Li, 2016)，如图1所示。这一分类为发音评估提供了清晰的区分依据。

### 2.1 发音错误

#### 语音错误

语音（音段）错误涉及对单个音素（如元音和辅音）的发音，主要包括三种错误类型：插入、删除和替换。这种现象可归因于多种因素，包括负面语言迁移、字母与音素转换错误以及对文本提示的误读(Meng 等, 2007b; Qian 等, 2010; Kartushina 与 Frauenfelder, 2014; Li 等, 2016a)。例如，阿拉伯语母语者可能难以区分 /p/ 与 /b/，因为阿拉伯语中不存在音素 /p/，因此诸如 /park/ 和 /bark/ 等动词对阿拉伯语母语者而言可能听起来很相似；同样，在西班牙语中不存在短元音，因此诸如 /eat/ 和 /it/ 等单词对西班牙语母语者而言可能听起来很相似。

#### 韵律错误

韵律特征涵盖影响整个单词或句子发音的各种要素，包括重音、节奏和语调。与韵律特征相关的语音错误涉及较大音段的产生。在确保语句可懂度方面，韵律特征发挥着尤为重要的作用 (Raux 与 Kawahara, 2002)。这一规律在声调语言中尤为突出 (Dahmen 等, 2023)，因为在这些语言中，音高变化可导致词语产生不同含义。韵律性错误通常具有语言依赖性，可按以下类别进行划分：重音（词汇与句子层面）、节奏及语调。

| Corpus                                           | Languages (L2)     | Native Language (L1)                                                        | Dur/Utt     | #Speakers | Reported SOTA Results / Relevant Studies                                                                                                                                              |
|--------------------------------------------------|--------------------|-----------------------------------------------------------------------------|-------------|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ISLE (Menzel et al., 2000) *                     | English            | German and Italian                                                          | 18/         | 46        | # PER:(Hosseini-Kivanani et al., 2021). Accent PCC: 68% (Rasipuram et al., 2015)                                                                                                      |
| ERJ (Minematsu et al., 2004) *                   | English            | Japanese                                                                    | /68,000     | 200       | # Utterance PCC (Luan et al., 2012). Word Intelligibility (Minematsu et al., 2011). Phoneme Errors (Ito et al., 2005)                                                                 |
| CU-CHLOE (Meng et al., 2007a)                    | English            | Cantonese and Mandarin                                                      | 34.6/18,139 | 210       | Phoneme F1-measure: 80.98% (Wu et al., 2021)                                                                                                                                          |
| EURONOUNCE (Cylwik et al., 2009)                 | Polish             | German                                                                      | /721        | 18        | # Utterance rythm (Wagner, 2014)                                                                                                                                                      |
| iCALL (Chen et al., 2015) +                      | Mandarin           | 24 countries                                                                | 142/90,841  | 305       | FAR: 8.65%, FRR: 3.09%: (Li et al., 2017). Tone Recognition: (Tong et al., 2015)                                                                                                      |
| SingaKids-Mandarin (?) +                         | Mandarin           | Singaporean (English)                                                       | 125/79,843  | 255       | PER: 28.51%. Tone Recognition (Tong et al., 2017)                                                                                                                                     |
| SHEFCE (Ng et al., 2017b) *                      | English, Cantonese | English, Cantonese                                                          | 25/         | 31        | Madarin syllable error rate: 17.3%, English PER: 34.5% (Ng et al., 2017a)                                                                                                             |
| VoisTUTOR (Yarra et al., 2019; Pal et al., 2022) | English            | Kannada, Malayalam, Telugu, Tamil, Hindi and Gujarati                       | 14/26,529   | 16        | Word Intelligibility Accuracy: 96.58% (Anand et al., 2023)                                                                                                                            |
| EpaDB (Vidal et al., 2019a) +                    | English            | Spanish                                                                     | /3,200      | 50        | (Sancinetti et al., 2022) reported Min-Cost per phoneme                                                                                                                               |
| SELL-CORPUS (Chen et al., 2019) *                | English            | Chinese                                                                     | 31.6/       | 389       | F1-score Accent Detection: Word-level 35%, Sentence-level 45% (Kyriakopoulos et al., 2020)                                                                                            |
| L2-ARCTIC (Zhao et al., 2018a) *                 | English            | Hindi, Korean, Mandarin, Spanish, and Arabic                                | 3.6/        | 24        | F1-score: 63.04% (Lin and Wang, 2022a)                                                                                                                                                |
| Speechocean762 (Zhang et al., 2021b) *           | English            | Chinese                                                                     | /5,000      | 250       | Phone PCC: 65.60% (Chao et al., 2022). Word Accuracy PCC: 59.80% (Chao et al., 2022). Word Stress PCC: 32.30% (Do et al., 2023). Sentence total score PCC: 79.60% (Chao et al., 2022) |
| LATIC (ZHANG, 2021) *                            | Mandarin           | Russian, Korean, French, and Arabic                                         | 4/2,579     | 4         | Sentence Accuracy PCC: 69.80% (Lin and Wang, 2023b)                                                                                                                                   |
| Arabic-CAPT (Algabri et al., 2022)               | Arabic             | India, Pakistan, Indonesia, Nepal, Afghanistan, Bangladesh, Nigeria, Uganda | 2.3/1,611   | 62        | F1-score 70.53% (Algabri et al., 2022)                                                                                                                                                |
| AraVoiceL2 (EL Kheir et al., 2023b)              | Arabic             | Turkey, Bangladesh, Nigeria, Indonesia, Malaysia                            | 5.5/7,062   | 11        | F1-score 60.00% (EL Kheir et al., 2023b)                                                                                                                                              |

Table 1: Widely used datasets. \* represent publicly available dataset, + is available on request, # relevant study, Dur: total duration in hours, Utt: total number of utterances, SOTA: is the notable reported state-of-the-art for each corpus. FAR: false acceptance rate, FRR: false rejection rate, PCC: pearson correlation coefficient with human scores

*Stress* is the emphasis placed on certain syllables in a word or sentence. It is articulated by increasing the loudness, duration, and pitch of the stressed syllable. It can be categorized as *lexical stress*, if the stress is placed on syllables within the word, or *sentence stress* if the stress is placed on words within sentences. Mandarin learners of English have contrastive stress at the word-level that is absent in Korean, Mandarin speakers can have an advantage over Korean speakers in stress

processing of English words (Wang, 2022).

*Rhythm* is the pattern of stressed and unstressed syllables in a word or sentence. A language can be classified as either stress-timed or syllable-timed (Ohata, 2004; Matthews, 2014). In stress-timed languages, the duration of stressed syllables tends to dominate the overall time required to complete a sentence. Conversely, in syllable-timed languages, each syllable receives an equal amount of time during production.

| 语料库                                    | 第二语言 (L2) | 母语 (L1)                             | Dur/Utt     | #演讲者 | 已报告的 SOTA 结果 / 相关研究                                                                                                            |
|----------------------------------------|-----------|-------------------------------------|-------------|------|--------------------------------------------------------------------------------------------------------------------------------|
| ISLE (Menzel 等, 2000)                  | 英语        | 德语与意大利语                             | 18/         | 46   | # PER: (Hosseini-Kivanani 等, 2021); Accent PCC: 68% (Rasipuram 等, 2015)                                                        |
| ERJ (Minematsu 等, 2004)                | 英语        | 日本人                                 | /68,000     | 200  | # 话语 PCC (Luan 等, 2012); 词理解度 (Minematsu 等, 2011); 音素错误率 (Ito 等, 2005)                                                         |
| CU-Chloe (Meng 等, 2007a)               | 英语        | 粤语与普通话                              | 34.6/18,139 | 210  | 音素 F1 值: 80.98% (Wu 等, 2021)                                                                                                   |
| Euronounce (Cyl-Wik 等, 2009)           | 抛光        | 德国的                                 | /721        | 18   | # 话语节奏 (Wagner, 2014)                                                                                                          |
| iCALL (Chen 等, 2015) +                 | 普通话       | 24个国家                               | 142/90,841  | 305  | FAR: 8.65%, FRR: 3.09%; (Li 等, 2017); 音调识别: (Tong 等, 2015)                                                                     |
| SingaKids- 中文版 (? ) +                  | 普通话       | 新加坡语 (英语)                           | 125/79,843  | 255  | 准确率: 28.51%。音色识别 (Tong 等, 2017)                                                                                                |
| SHEFCE (Ng 等, 2017 b)                  | 英语、粤语     | 英语、粤语                               | 25/         | 31   | Madarin 音节错误率: 17.3%, 英语词频 (PER): 34.5% (Ng 等, 2017a)                                                                          |
| VoisTUTOR (Yarra 等, 2019; Pal 等, 2022) | 英语        | 卡纳达语、马拉雅拉姆语、泰卢固语、泰米尔语、印地语及古吉拉特语     | 14/26,529   | 16   | 词汇理解度准确率: 96.58% (Anand 等, 2023)                                                                                               |
| EpaDB (Vidal 等, 2019a) +               | 英语        | 西班牙的                                | /3,200      | 50   | (Sancinetti 等, 2022) 报道了每个音素的最小成本                                                                                              |
| SELL-corpus (Chen 等, 2019)             | 英语        | 中国人                                 | 31.6/       | 389  | F1 分数重音检测: 词级 35%, 句子级 45% (Kyriakopou 等, 2020)                                                                                |
| L2- 北极地区 (Zhao 等, 2018a)               | 英语        | 印地语、韩语、普通话、西班牙语和阿拉伯语                | 3.6/        | 24   | F1 分数: 63.04% (Lin 与 Wang, 2022 a)                                                                                             |
| Speechocean762 (Zhang 等, 2021b)        | 英语        | 中国人                                 | /5,000      | 250  | 手机 PCC: 65.60% (Chao 等, 2022)。单词准确性 PCC: 59.80% (Chao 等, 2022); 单词重音 PCC: 32.30% (Do 等, 2023); 句子总分 PCC: 79.60% (Chao 等, 2022) |
| LATIC (张, 2021)                        | 普通话       | 俄语、韩语、法语和阿拉伯语                       | 4/2,579     | 4    | 句子准确率 PCC: 69.80% (Lin 与 Wang, 2023b)                                                                                          |
| Arabic-CAPT (Algabri 等, 2022)          | 阿拉伯的      | 印度、巴基斯坦、印度尼西亚、尼泊尔、阿富汗、孟加拉国、尼日利亚、乌干达 | 2.3/1,611   | 62   | F1 分数: 70.53% (Algabri 等, 2022)                                                                                                |
| AraVoiceL2 (EL Kheir 等, 2023b)         | 阿拉伯的      | 土耳其, 尼日利亚、孟加拉国、印度尼西亚、马来西亚           | 5.5/7,062   | 11   | F1 分数为 60.00% (EL Kheir 等, 2023 b)                                                                                             |

表1: 广泛使用的数据集。\* 表示公开可用的数据集, + 表示可根据需求提供, # 表示相关研究。  
Dur: 总时长 (单位: 小时); Utt: 总话语数; SOTA: 各指标中公认的最先进水平  
语料库。FAR: 误接受率; FRR: 误拒绝率; PCC: 与人工评分之间的皮尔逊相关系数

重音是指在单词或句子中对特定音节所施加的强调; 其表现为增加被重读音节的响度、时长和音高。若重音落在单词内部的音节上, 则可归类为*词汇重音*; 若重音落在句子中的单词上, 则可归类为*句子重音*。学习英语的汉语母语者在词级层面存在对比性重音现象, 而韩语母语者则不具备这种现象; 因此, 在处理英语单词的重音时, 汉语母语者可能比

韩语母语者更具优势 (Wang, 2022)。  
*节拍*是指单词或句子中重读音节与非重读音节的排列模式。语言可划分为重音时制语言或音节时制语言 (Ohata, 2004; Matthews, 2014)。在重音时制语言中, 重读音节的时长往往占整个句子生成所需总时长的主导地位; 相反, 在音节时制语言中, 每个音节在语音生成过程中所占的时间长度相等。

*Intonation* refers to the melodic pattern and pitch variations in speech. L2 learners of Vietnamese and Mandarin Chinese encounter significant difficulty in acquiring distinct tones, particularly if their native language lacks tonality. Such tonal languages rely on different pitch patterns to convey distinct meanings, making it challenging for learners to accurately grasp and reproduce these tonal variations (Nguyen et al., 2014; Chen et al., 2015).

## 2.2 Pronunciation Constructs

The motivation behind mastering L2 pronunciation is to communicate properly in the target language. Most of the time, these successes are measured using three pronunciation constructs (Uchihara, 2022) – **Intelligibility**, **Comprehensibility**, and **Accentedness**. These are perceived measures, that are partially independent with overlapping features.

**Intelligibility** can be defined using the accuracy of the sound, word, and utterance itself along with utterance-level completeness (Abercrombie, 1949; Gooch et al., 2016). Accuracy refers where the learner pronounces each phoneme, or word in the utterance correctly. In contrast, completeness measures the percentage of words pronounced compared to the total number of words.

**Comprehensibility**, on the other hand, is defined based on the perceived ease or difficulty that listeners experience when understanding L2 speech. Fluency, defined by the smoothness of pronunciation and correct usage of pauses (Zhang et al., 2021b), is observed to be one of the key factors that determine the level of comprehensibility, along with good linguistic-knowledge and discourse-level organization (Trofimovich and Isaacs, 2012; Saito et al., 2016).

Among the three constructs, **accentedness**, which is defined as “listeners’ perceptions of the degree to which L2 speech is influenced by their native language and/or colored by other non-native features” (Saito et al., 2016). It is often confused with both comprehensibility and intelligibility, influencing pronunciation assessment. The accent is an inherent trait that defines a person’s identity and is one of the first things that a listener notices. It is often observed that most of the unintelligible speech is identified as highly accented whereas highly accented speech is not always unintelligible (Derwing and Munro, 1997; Kang et al., 2018; Munro and Derwing, 1995). Thus accents complicate fine-grained pronunciation assessment as it is harder to pinpoint (supra-)segment-level error.

## 3 Datasets

Obtaining datasets for pronunciation assessment is often challenging and expensive. Most of the available research work focused on private data, leaving only a handful of publicly accessible data to the research community. Table 1 provides an overview of available datasets, indicating English as a popular choice for the target language. Within this handful of datasets, a few datasets include phonetic/segmental-level transcription and even fewer provide manually rated word and sentence-level prosodic features, fluency along with overall proficiency scores offering insights to learner’s L2 speech intelligibility and comprehensiveness (Arvaniti and Baltazani, 2000; ?; Cole et al., 2017; Zhang et al., 2021b). More details on datasets and annotation are in Appendix A and B respectively.

## 4 Research Avenues

In this section, we will delve into diverse approaches, old, revised, and current methodologies used for pronunciation modeling of both segmental and supra-segmental features, as illustrated in Figure 2 and Figure 3.

### 4.1 Classification based on Acoustic Phonetics

Classifier-based approaches explored both segmental and prosodic aspects of pronunciation. *Segmental* approaches involve the use of classifiers targeting specific phoneme pair errors, utilizing different acoustic features such as Mel-frequency cepstral coefficients (MFCCs) along with its first and second derivative, energy, zero-cross, and spectral features (Van Doremalen et al., 2009; Huang et al., 2020), with different techniques such as Linear Discriminant Analysis (LDA) (Truong et al., 2004; Strik et al., 2009), decision trees (Strik et al., 2009). *Prosodic* approaches focus on detecting lexical stress and tones, utilizing features such as energy, pitch, duration, and spectral characteristics, with classifiers like Gaussian mixture models (GMMs) (Ferrer et al., 2015), support vector machines (SVMs) (Chen and Wang, 2010; Shahin et al., 2016), and deep neural network (DNNs) (Shahin et al., 2016), and multi-distribution DNNs (Li et al., 2018a).

### 4.2 Extended Recognition Network (ERN)

ERNs are neural networks used in automatic speech recognition to capture broader contextual information, they leverage enhanced lexicons in combination with ASR systems. They cover canonical tran-

语调是指言语中的旋律模式与音高变化。学习越南语或普通话的第二语言学习者在习得 distinct 语调时往往面临较大困难，尤其是当其母语不具有声调特征时。这类声调语言依赖不同的音高模式来传递 distinct 意义，因此学习者要准确掌握并再现这些声调变化颇具挑战性（Nguyen 等，2014；Chen 等，2015）。

## 2.2 发音结构

掌握第二语言（L2）发音的动机在于能够用目标语言进行恰当的交流。在大多数情况下，这些学习成效是通过三种发音评估指标来衡量的（Uchihara，2022）。

– **语义清晰度、语义可理解性与重音度**。这些均为感知度量指标，彼此部分独立且具有重叠的特征。

**可懂度**可定义为声音、单词及话语本身的准确度，结合话语层面的完整性（Abercrombie，1949；Gooch 等，2016）。准确度指学习者对话语中每个音素或单词的发音是否正确；而完整性则衡量了实际发音单词数占总单词数的百分比。

**可理解性**则是基于听者在理解第二语言（L2）语音时所感受到的易理解度或难理解度来定义的。流利度——即发音的流畅性与停顿的正确运用（Zhang 等，2021b）——被观察为决定可理解性水平的关键因素之一，此外，良好的语言知识与话语层面的组织结构亦是影响因素（Trofimovich 与 Isaacs，2012；Saito 等，2016）。

在三个构念中，**重音度**被定义为“听者对第二语言语音受到其母语影响和/或受到其他非母语特征影响程度的感知”（Saito 等，2016）。该构念常与可理解度及清晰度相混淆，从而对发音评估产生影响。重音是一种定义个人身份的固有特征，也是听者最先注意到的特征之一。通常观察到，大多数难以理解的语音被归类为具有高重音度，而具有高重音度的语音并不总是难以理解（Derwing 与 Munro，1997；Kang 等，2018；Munro 与 Derwing，1995）。因此，重音使得细粒度的发音评估变得复杂，因为要精准定位（超）音段级错误更为困难。

## 3 数据集

获取用于发音评估的数据集往往具有挑战性且成本高昂。目前大多数现有研究工作均聚焦于私有数据，导致研究界仅能使用少数公开可获取的数据集。表1对现有数据集进行了概述，其中英语被列为常用的目标语言。在这些少数数据集中，部分数据集包含音素/分段级转写数据，而提供人工标注的词级及句级韵律特征、流利度以及整体熟练度评分的数据集则更为稀少——这些数据可为学习者第二语言的语音可懂度与全面性提供深入见解（Arvaniti 与 Baltazani，2000；？；Cole 等，2017；Zhang 等，2021b）。关于数据集与标注的更多详细信息，分别见附录A与B。

## 4 研究方向

在本节中，我们将深入探讨用于音段及超音段特征发音建模的各种方法——包括传统方法、修订后的方法以及当前采用的方法，如图2和图3所示。

### 4.1 基于声学语音学的分类方法

基于分类器的方法同时探究了发音中的片段化与韵律学层面。片段化方法采用针对特定音素对错误的分类器，利用多种声学特征（如梅尔频率倒谱系数（MFCCs）及其一阶和二阶导数、能量、零交叉点及频谱特征Van Doremalen 等，2009；Huang 等，2020），并结合线性判别分析（LDA）（Truong 等，2004；Strik 等，2009）、决策树（Strik 等，2009）等不同技术。韵律学方法侧重于检测词汇重音与声调，利用能量、音高、时长及频谱特征等特征，并采用高斯混合模型（GMMs）（Ferrer 等，2015）、支持向量机（SVMs）（Chen 和 Wang，2010；Shahin 等，2016）以及深度学习等分类器。深度神经网络（DNNs）（Shahin 等，2016）以及多分布深度神经网络（Li 等，2018a）。

### 4.2 扩展认可网络（ERN）

ERNs 是用于自动语音识别的神经网络，旨在捕获更广泛的上下文信息；它们将增强型词典与 ASR 系统相结合。这些网络不仅涵盖标准

scriptions as well as error patterns, enabling the detection of mispronunciations beyond standard transcriptions (Meng et al., 2007b; Ronen et al., 1997; Qian et al., 2010; Li et al., 2016b). However, ERNs often depend on experts or hand-crafted error patterns, which are typically derived from non-native speech transcriptions as illustrated in (Lo et al., 2010) which makes it language-dependent approach and may limit their generalizability when dealing with unknown languages.

### 4.3 Likelihood-based Scoring and GOP

The initial likelihood-based MD algorithms aim to detect errors at the phoneme level using pre-trained HMM-GMM ASR models. Notably, Kim et al. (1997) introduced a set of three HMM-based scores, including likelihood scores, log posterior scores, and segment-duration-based scores. Among these three, the log-based posterior scores are widely adopted due to their high correlation with human scores, and are also used to calculate the popular ‘goodness of pronunciation’ (GOP) measure. The GMM-HMM based GOP scores can be defined by the Equation 1.

$$GOP(p) = P(p|O) = \frac{p(O|p) P(p)}{\sum_q p(O|q) P(q)} \quad (1)$$

$O$  denotes a sequence of acoustic features,  $p$  stands for the target phone, and  $Q$  represents the set of phones.

These scores are further improved using forced alignment framework (Kawai and Hirose, 1998). More details are presented in Witt and Young (2000).

### 4.4 Reformulations of GOP

To enhance further the effectiveness of GOP scoring, (Zhang et al., 2008) are first to propose a log-posterior normalized GOP defined as:

$$GOP_r(p) = \left| \frac{p(o_t|p)P(p)}{\max_q p(o_t|q)} \right| \quad (2)$$

Building upon this, Wang and Lee (2012) adopted the GOP formulation and incorporate error pattern detectors for phoneme mispronunciation diagnosis tasks. With the emergence of DNN in the field of ASR, Hu et al. (2013, 2015a,b) demonstrated that using a DNN-HMM ASR for GOP yields improved correlation scores surpassing GMM-HMM based GOP. The GOP and its reformulation represent a significant milestone. It leverages pre-trained acoustic models on the target language without the

necessitating of speaker’s L1 knowledge. Furthermore, it offers the advantage of being computationally efficient to calculate. However, these scores lack context-aware information that is crucial for accurate pronunciation analysis. To overcome this, Sudhakara et al. (2019) presented a context-aware GOP formulation by adding phoneme state transition probabilities (STP) extracted from HMM model to the GOP score calculation. Furthermore, Shi et al. (2020) proposed a context-dependent GOP, incorporating a phoneme duration factor  $\alpha_i$ , and phonemes transition factor  $\tau$ . The formulated GOP score combines all the contextual scores as illustrated in Equation 3.

$$\begin{aligned} E_t &= - \sum p(q|O) \log(p(q|O)) \\ \tau(p) &= \sum_t \frac{\frac{1}{E_t}}{\sum_{t'} \frac{1}{E_{t'}}} \log(p(q|O)) \\ GOP(p) &= (1 - \alpha_i) * \tau(p) \end{aligned} \quad (3)$$

For *sentence accuracy* evaluation, one common approach is to calculate the average GOP scores across phonemes (Kim et al., 1997; Sudhakara et al., 2019). However, relying solely on averaging GOP scores at the phoneme level is limited. A recent approach in (Sheoran et al., 2023) proposed a combination of phone feature score and audio pitch comparison using dynamic time warping (DTW) with an ideal pronounced speech, as a score to assess prosodic, fluency, completeness, and accuracy at the sentence level. Inspired by GOP, Tong et al. (2015) proposed Goodness of Tone (GOT) based on posterior probabilities of tonal phones.

While efforts have been made to improve the GOP formulation, it is important to acknowledge that the GOP score still has limitations, specifically in its ability to identify specific types of mispronunciation errors (deletion, insertion, or substitution), and it also demonstrates a degree of dependency on the language of the acoustic model.

### 4.5 End-to-End Modeling

In the new era of DNNs and Transformers, there is a significant exploration by researchers in leveraging the power of these models and training end-to-end pronunciation systems. Li et al. (2017) introduced LSTM mispronunciation detector leveraging phone-level posteriors, time boundary information, and posterior extracted from trained DNNs models on the classification of phonetic attributes (place, manner, aspiration, and voicing). In contrast, Kyriakopoulos et al. (2018) introduced

转写形式，还涵盖错误模式，从而能够检测到超越标准转写形式的错误发音（Meng 等，2007b; Ronen 等，1997; Qian 等，2010; Li 等，2016b）。然而，ERNS 通常依赖于专家标注或人工构建的错误模式，这些模式通常源自非母语语音转写（如Lo等，2010所示），这使得其属于语言依赖性方法，在处理未知语言时可能限制其泛化能力。

### 4.3 基于似然函数的评分法与 GOP

基于似然性的初始MD算法旨在利用预训练的 HMM- GMM ASR模型在音素层面检测错误。值得注意的是，Kim 等人(1997)提出了一套包含三种基于HMM的评分方法，包括似然评分、对数后验评分以及基于片段时长的评分。在这三种方法中，基于对数的后验评分因其与人工评分具有高度相关性而被广泛采用，同时也用于计算流行的“发音质量”（GOP）度量指标。基于 GMM -HMM的GOP评分可由公式1定义。

$$GOP(p) = P(p|O) = \frac{p(O|p) P(p)}{\sum_q p(O|q) P(q)} \quad (1)$$

$O$ 表示一串声学特征序列， $p$ 代表目标音素，而 $Q$ 表示音素集合。

这些得分通过强制对齐框架（Kawai 与 Hirose，1998）得到了进一步提升。更多细节请参阅Witt 与 Young(2000)。

### 4.4 GOP配方调整

为进一步提升 GOP 评分法的有效性，(Zhang 等，2008)首次提出了一个对数后验归一化 GOP，其定义为：

$$GOP_r(p) = \left| \frac{p(o_t|p)P(p)}{\max_q p(o_t|q)} \right| \quad (2)$$

在此基础上，Wang 与 Lee(2012)采用了 GOP 模型，并为音素误读诊断任务整合了错误模式检测器。随着 DNN 在自动语音识别（ASR）领域的兴起，Hu 等人（2013、2015a, b）证明了采用 DNN -HMM ASR 进行 GOP 计算可获得比基于 GMM -HMM 的 GOP 更高的相关性得分。GOP 及其改进版本代表了一个重要的里程碑——该模型无需依赖说话者的第一语

言知识，即可利用针对目标语言预训练的声学模型；此外，其计算效率较高。然而，这些得分缺乏对上下文信息的考量，而这种信息对于准确的发音分析至关重要。为解决这一问题，Sudhakara 等人(2019)提出了一种上下文感知的 GOP 模型，方法是将从 HMM 模型中提取的音素状态转移概率（STP）加入到 GOP 得分计算中。此外，Shi 等人（2020）提出了一种上下文依赖的 GOP，包含音素时长因子 $\alpha_i$ 与音素转移因子 $\tau$ 。所构建的 GOP 评分将所有上下文评分综合起来，如公式3所示。

$$\begin{aligned} E_t &= - \sum p(q|O) \log(p(q|O)) \\ \tau(p) &= \sum_t \frac{\frac{1}{E_t}}{\sum_{t'} \frac{1}{E_{t'}}} \log(p(q|O)) \\ GOP(p) &= (1 - \alpha_i) * \tau(p) \end{aligned} \quad (3)$$

对于句子准确度评估，一种常用方法是计算各音素层面的平均GOP分数（Kim 等，1997; Sudhakara 等，2019）。然而，仅依赖音素层面的GOP分数平均值存在局限性。近期一项研究（Sheoran 等，2023）提出将音素特征分数与基于动态时间规整（DTW）技术、以理想清晰语音为参照的音频音高比较相结合，作为评估句子层面韵律性、流畅度、完整性和准确度的评分指标。受GOP方法启发，Tong 等（2015）提出了基于声调音素后验概率的声调质量度量（GOT）。

尽管已对 GOP 评分公式进行了改进，但必须承认，该评分系统仍存在局限性——尤其是在识别特定类型发音错误（删除、插入或替换）方面的能力尚有不足；此外，该系统在一定程度上仍依赖于所采用声学模型的语言。

### 4.5 端到端建模

在深度神经网络（DNNs）与Transformer模型主导的新时代，研究人员积极探索如何充分发挥这些模型的潜力，并对端到端的发音识别系统进行训练。Li 等人（2017）提出了一个名为 LSTM 的误读检测器，该方法利用音素级后验概率、时间边界信息以及从经过训练的DNN模型中提取的后验概率，对语音属性（发音部位、发音方式、送气性和浊音性）进行分类。相比之下，Kyriakopoulos 等人（2018）提出了

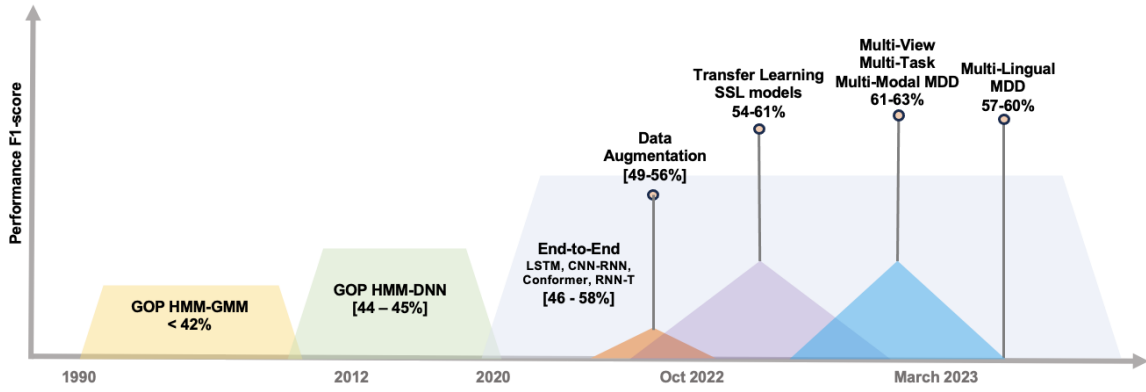


Figure 2: Overview of the performance of different phonetic pronunciation detection models on L2-ARCTIC

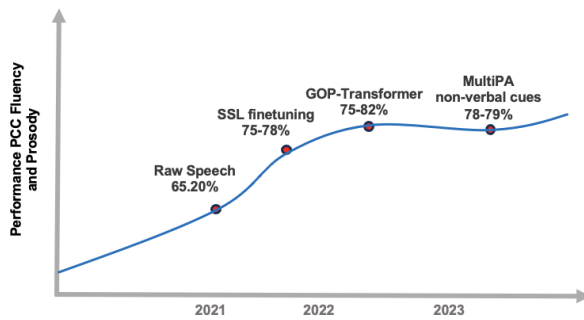


Figure 3: Overview of the performance of fluency and prosody assessment models on Speechocean762

a siamese network with BiLSTM for pronunciation scoring by extracting distance metrics between phone instances from audio frames. A notable approach presented in Leung et al. (2019) introduced a CNN-RNN-CTC model for phoneme mispronunciation detection without any alignment component dependency. Subsequently, Feng et al. (2020) incorporated character embeddings to enhance CNN-RNN model. Furthermore, Ye et al. (2022) enhanced the later model using a triplet of features consisting of acoustic, phonetic, and linguistic embeddings. Subsequently, GOP features extracted from pre-trained ASR are enhanced using a Transformer encoder to predict a range of scores of prosodic and segmental scores (Gong et al., 2022), or using additional SSL representation features, energy, and duration within the same architecture (Chao et al., 2022), or using Conformer encoder explored in (Fan et al., 2023). Moreover, PEPANET is also a transformer-based mispronunciation model, but can jointly model the dictation process and the alignment process, and it provides corresponding diagnostic feedback (Yan

et al., 2023a). A subsequent improvement of PEPANET uses knowledge about phone-level articulation traits with a graph convolutional network (GCN) to obtain more discriminative phonetic embeddings (Yan et al., 2023b). Recently, Zhang et al. (2023) proposed recurrent neural network transducer RNN-T for L2 phoneme sequence prediction along with an extended phoneme set and weakly supervised training strategy to differentiate similar-sounding phonemes from different languages.

Several approaches have also been proposed for *supra-segmental features scoring*. Yu et al. (2015), proposed a new approach where traditional time-aggregated features are replaced with time-sequence features, such as pitch, to preserve more information without requiring manual feature engineering, a BiLSTM model is proposed for fluency predictions. Tao et al. (2016); Chen et al. (2018), studied different DNNs models such as CNN, BiLSTM, Attention BiLSTM to predict the fluency and prosodic scoring. (Lin and Wang, 2021) utilized deep features directly from the acoustic model instead of relying on complex feature computations like GOP scores with a scoring module, incorporating a self-attention mechanism, which is designed to model human sentence scoring. More recently, (Zhu et al., 2023) proposed BiLSTM model trained to predict the intelligibility score of a given phoneme or word segment using an annotated intelligibility L2 speech using shadowing.

Towards *lexical stress* detection, several methods have been proposed to improve accuracy and performance. Ruan et al. (2019) proposed a sequence-to-sequence approach using the Transformer model upon the need for long-distance contextual information to predict phoneme sequence with stress marks. Furthermore, Korzekwa et al. (2020a) intro-

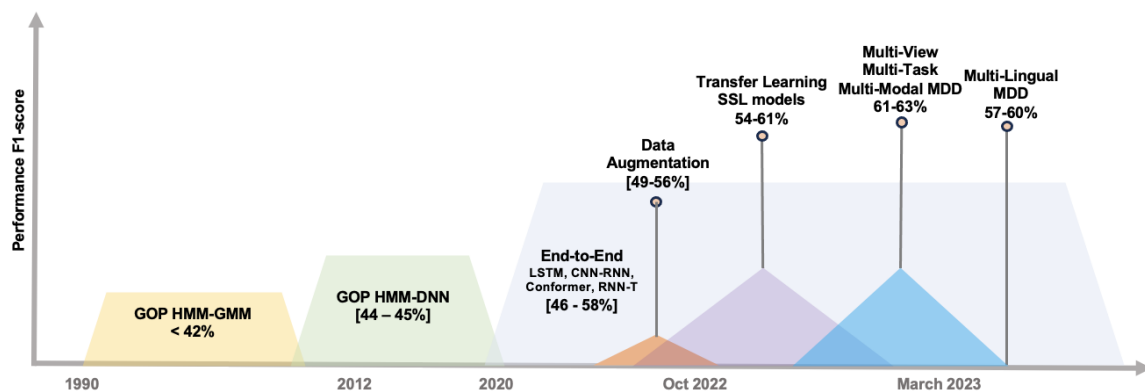


图 2: 不同语音发音检测模型在 L2-北极地区数据集上的性能概览

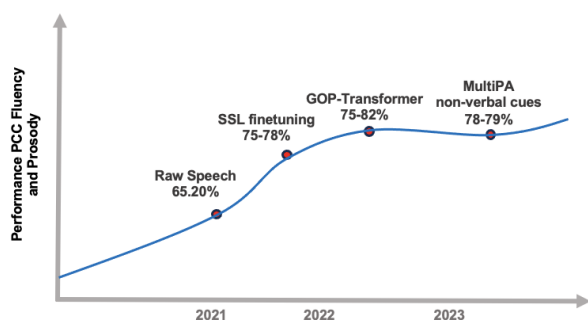


图 3: Fluency 和 Prosody 评估模型在 SpeechOcean762 数据集上的性能概览

一种采用 BiLSTM 的孪生网络模型，通过提取音频帧中音素实例之间的距离度量来进行发音评分。Leung 等人 (2019) 提出了一种 CNN-RNN-CTC 模型用于音素误读检测，该模型无需依赖任何对齐组件。随后，Feng 等人 (2020) 引入了字符嵌入技术对 CNN-RNN 模型进行增强。此外，Ye 等人 (2022) 利用由声学、语音学和语言学嵌入组成的三元特征对上述模型进行了改进。随后，从预训练的 ASR 模型中提取的 GOP 特征通过 Transformer 编码器进行增强，以预测一系列韵律学与片段化评分 (Gong 等人, 2022)；或在相同架构中结合额外的 SSL 表征特征、能量及时长进行处理 (Chao 等人, 2022)；亦或采用 Fan 等人 (2023) 所探讨的 Conformer 编码器进行处理。此外，Peppanet 同样是一个基于 Transformer 的发音错误识别模型，但它能够联合建模听写过程与对齐过程，并提供相应的诊断反

馈 (Yan 等, 2023a)。随后对 Peppanet 的改进方法利用音素级发音特征知识结合图卷积网络 (GCN) 来获取更具区分度的音素嵌入表示 (Yan 等, 2023b)。近期，Zhang 等 (2023) 提出了循环神经网络转换器 RNN-T 用于第二语言音素序列预测，并结合扩展音素集与弱监督训练策略，以区分来自不同语言的发音相似音素。

此外，还提出了多种用于超段级特征评分的方法。Yu 等人 (2015) 提出了一种新方法，将传统的时序聚合特征替换为时序特征（如音高），从而在无需人工特征工程的情况下保留更多信息；同时，他们提出了一种 BiLSTM 模型用于流利度预测。Tao 等人 (2016)；Chen 等人 (2018) 研究了多种深度神经网络模型（如 CNN、BiLSTM、Attention BiLSTM）用于预测流利度和韵律评分。Lin 与 Wang (2021) 直接利用来自声学模型的深度特征，而非依赖如通过评分模块计算 GOP 分数这类复杂的特征计算方法，并引入了自注意力机制，该机制旨在模拟人类对句子的评分方式。近期，Zhu 等人 (2023) 提出了一种 BiLSTM 模型，该模型通过使用带有标注的、基于影子法的第二语言语音数据来训练，用于预测给定音素或词段的可懂度评分。

在词汇重音检测领域，已有多种方法被提出以提高检测的准确性和性能。Ruan 等人 (2019) 针对需要利用长距离上下文信息来预测带重音标记的音素序列这一需求，提出了一种基于 Transformer 模型的序列到序列方法。此外，Korzekwa 等人 (2020a) 引入了一种基于注

duced an attention-based neural network focusing on the automatic extraction of syllable-level features that significantly improves the detection of lexical stress errors.

*Tone classification* has received significant attention in Mandarin language learning due to the crucial role that tones play in Mandarin Chinese. To address the challenge several methods have been proposed. One approach involves training a DNN to classify speech frames into six tone classes (Ryant et al., 2014). Inspired by this, DNNs have been used to map combined cepstral and tonal features to frame-level tone posteriors. These tone posteriors are then fed into tone verifiers to assess the correctness of tone pronunciation (Lin et al., 2018; Li et al., 2018b). Another study utilizes CNN to classify syllables into four Mandarin tones (Chen et al., 2016a). Similarly, ToneNet, a CNN-based network is introduced for Chinese syllable tone classification using mel-spectrogram as a feature representation (Gao et al., 2019). Additionally, a BiLSTM model is proposed as an alternative to capture long-term dependencies in acoustic and prosodic features for tone classification (Li et al., 2019).

#### 4.6 Self-Supervised Models

Motivated by the recent success of self-supervised learning methods (Baeovski et al., 2020; Hsu et al., 2021; Chen et al., 2022; Mohamed et al., 2022) in speech recognition and related downstream tasks such as emotion recognition, speaker verification, and language identification (Chen and Rudnicky, 2023; Fan et al., 2020), self-supervised approaches is employed also in this field.

Xu et al. (2021) explored finetuning wav2vec 2.0 on frame-level L2 phoneme prediction. A pre-trained HMM-DNN ASR is used to extract time force-alignment. To overcome the dependency on time alignment, Peng et al. (2021) propose a CTC-based wav2vec 2.0 to predict L2 phonemes sequences. Building upon this work, Yang et al. (2022) propose an approach that leverages unlabeled L2 speech using momentum pseudo-labeling. In a contrasting approach, (Lin and Wang, 2022b) combined wav2vec 2.0 features and phoneme text embeddings in a jointly learning framework to predict frame-level phoneme sequence and detect boundaries. Recently, EL Kheir et al. (2023a) explored the multi-view representation utilizing mono- and multilingual wav2vec 2.0 encoders to capture different aspects of speech production and

leveraging articulatory features as auxiliary tasks to phoneme sequence prediction. Furthermore, Kheir et al. (2023b) introduces a novel L1-aware multilingual, L1-MultiMDD, architecture for addressing mispronunciation in multilingual settings encompassing Arabic, English, and Mandarin using wav2vec-large pre-trained model as the acoustic encoder. L1-MultiMDD is enriched with L1-aware speech representation, allowing it to understand the nuances of each speaker’s native language.

SSL models have proven to be effective in predicting fluency and prosodic scores assigned by human annotators. Kim et al. (2022); Lin and Wang (2023a); Yang et al. (2022) fine-tuned wav2vec 2.0 and Hubert to predict prosodic and fluency scores.

Similarly, another research conducted in (Lin and Wang, 2023a) jointly predicts L2 phoneme sequence using CTC loss, and predicts prosodic scores using fused acoustic representations with phoneme embeddings. Subsequently Lin and Wang (2023b) introduced a fusion of language embedding, representation features and build a unified framework for multi-lingual prosodic scoring. Recently, Chao et al. (2022); Kheir et al. (2023a); Chen et al. (2023), enriched latent speech extracted from SSL models with handcrafted frame- and utterance-level non-verbal paralinguistic cues such as duration, and energy for modeling Fluency and Prosody scores.

#### 4.7 Unsupervised Approaches

It is important to note that the aforementioned approaches for studying mispronunciation detection typically involve the need for expert knowledge, laborious manual labeling, or dependable ASR results, all of which come with significant costs. In contrast, recent years have witnessed considerable endeavors in unsupervised acoustic pattern discovery, yielding sub-optimal outcomes. Lee and Glass (2012) initially investigated a comparison-based approach that analyzes the extent of misalignment between a student’s speech and a teacher’s speech. In subsequent studies Lee and Glass (2015); Lee et al. (2016), explored the discovery of mispronunciation errors by analyzing the acoustic similarities across individual learners’ utterances, with a proposed n-best filtering method to resolve ambiguous error candidate hypotheses derived from acoustic similarity clustering. Furthermore, Mao et al. (2018) proposed *k*-means clustering on phoneme-based phonemic posterior-grams (PPGs) to expand the phoneme set in L2 speech. More recently, Sini et al.

意力机制的神经网络，该网络专注于自动提取音节级特征，从而显著提升了词汇重音错误检测的性能。

声调分类在普通话语言学习中受到了广泛关注，这是由于声调在普通话中发挥着至关重要的作用。为应对这一挑战，学界提出了多种方法。其中一种方法是训练一个 DNN 模型将语音帧分类为六个声调类别 (Ryant 等, 2014)。受此启发，研究者们利用深度神经网络 (DNN) 将结合了倒谱和声调特征的信息映射到帧级声调后验概率上；随后将这些声调后验概率输入声调验证器中，以评估声调发音的正确性 (Lin 等, 2018; Li 等, 2018b)。另一项研究采用卷积神经网络 (CNN) 将音节分类为四种普通话声调 (Chen 等, 2016a)。类似地，研究者提出了一个基于 CNN 的 ToneNet 网络，利用梅尔频谱图作为特征表示来实现中文音节声调分类 (Gao 等, 2019)。此外，还提出了一种 BiLSTM 模型作为替代方案，用于捕捉声学和谐律特征中的长程依赖关系以进行声调分类 (Li 等, 2019)。

#### 4.6 自监督模型

受近期自监督学习方法 (Baevski 等, 2020; Hsu 等, 2021; Chen 等, 2022; Mohamed 等, 2022) 在语音识别及情感识别、说话人验证和语言识别等相关下游任务 (Chen 与 Rudnicky, 2023; Fan 等, 2020) 中取得的成功启发，自监督方法亦被应用于该领域。

Xu 等 (2021) 研究了在帧级 L2 语音素预测任务上对 wav2vec 2.0 进行微调的方法。该研究采用预训练的 HMM-DNN 语音识别模型来提取时间力对齐信息。为克服对时间对齐的依赖性，Peng 等 (2021) 提出了一种基于 CTC 的 wav2vec 2.0 模型用于预测 L2 语音素序列。基于这项工作，Yang 等 (2022) 提出了一种利用无标签 L2 语音数据并通过动量伪标注技术进行建模的方法。在另一种对比性方法中，(Lin 与 Wang, 2022b) 将 wav2vec 2.0 特征与语音素文本嵌入相结合，构建了一个联合学习框架，用于预测帧级语音素序列并检测语音边界。近期，EL Kheir 等 (2023a) 探索了多视图表征方法，利用单语及多语 wav2vec 2.0 编码器来捕捉语音产生

过程中的不同方面，并将发音特征作为辅助任务用于语音素序列预测。此外，Kheir 等 (2023b) 提出了一种新型的 L1 感知多语言 L1-MultiMDD 是一款用于解决多语言环境（涵盖阿拉伯语、英语和普通话）中发音错误问题的架构，其采用 wav2vec-large 预训练模型作为声学编码器。L1-MultiMDD 集成了 L1 感知语音表征技术，使其能够理解每位说话者母语中的细微差别。

SSL 模型已被证明在预测人类标注者所赋予的流利度与韵律评分方面效果显著。Kim 等人 (2022); Lin 与 Wang (2023a); Yang 等人 (2022) 对 wav2vec 2.0 和 Hubert 模型进行微调，用于预测韵律与流利度评分。类似地，另一项研究 (Lin 与 Wang, 2023a) 采用 CTC 损失函数联合预测第二语言音素序列，并利用融合的声学表示与音素嵌入来预测韵律评分。随后，Lin 与 Wang (2023b) 引入了语言嵌入与表征特征的融合方法，构建了一个用于多语言韵律评分的统一框架。近期，Chao 等人 (2022); Kheir 等人 (2023a);

Chen 等 (2023)，从基于自监督学习 (SSL) 的模型中提取的富集潜在语音，该语音结合了人工设计的帧级与话语级非言语副语言线索（如时长和能量），用于建模流畅度与韵律评分。

#### 4.7 无监督学习方法

需要特别注意的是，上述用于研究错误发音检测的方法通常需要依赖专家知识、繁琐的手动标注或可靠的 ASR 结果，而这些方法均伴随较高的成本。相比之下，近年来在无监督声学模式发现领域开展了大量研究，但所得结果尚非最优。Lee 与 Glass (2012) 最初探索了一种基于比较的方法，用于分析学生语音与教师语音之间错位程度；在后续研究中，Lee 与 Glass (2015); Lee 等 (2016) 通过分析个体学习者话语间的声学相似性来探索错误发音的发现，并提出一种 n-best 过滤方法，用于解决从声学相似性聚类中得出的模糊错误候选假设问题。此外，Mao 等 (2018) 提出了基于音素的音素后验图的  $k$  均值聚类方法。(PPGs) 用于扩展第二语言语音中的音素集。近期，Sini 等

(2023) introduced a weighted DTW alignment as an alternative to the GOP algorithm for predicting probabilities and the sequence of target phonemes. Their proposed method achieves comparable results to the GOP scoring algorithm, likewise Anand et al. (2023) explored alignment distance between wav2vec 2.0 representations of teacher and learner speech using DTW, to distinguish between intelligible and unintelligible speech.

#### 4.8 Data Augmentation

Two major challenges in this field are L2 data scarcity and the imbalanced distribution of negative classes (mispronunciation). To address these challenges, researchers have opted for data augmentation techniques that are proven to be quite effective in pronunciation assessment. Such methods employed strategies like altering the canonical text by introducing mismatched phoneme pairs while preserving the original word-level speech (Fu et al., 2021). Additionally, a mixup technique is utilized in the feature space, leveraging phone-level GOP pooling to construct word-level training data (Fu et al., 2022). Furthermore, the error distance of the clustered SSL model embeddings are employed to substitute the phoneme sound with a similar sound (Zhang et al., 2022b). These latter approaches depend on the reuse of existing information rather than generating novel instances of mispronunciations. In (Fernandez et al., 2017), voice transformations in pitch, vocal-tract, vocal-source characteristics to generate new samples. Furthermore, L2-GEN can synthesize realistic L2 phoneme sequences by building a novel Seq2Seq phoneme paraphrasing model (Zhang et al., 2022a). Korzekwa et al. (2020b) proposed an augmentation technique by generating incorrectly stressed words using Neural TTS. Furthermore, Korzekwa et al. (2022) provided an overview of mispronunciation error generation using three methods, phoneme-2-phoneme P2P relies on perturbing phonetic transcription for the corresponding speech audio, text-2-speech create speech signals that match the synthetic mispronunciations, and speech-2-speech S2S to simulate a different aspect of prosodic nature of speech. Recently, SpeechBlender (EL Kheir et al., 2023b) framework is introduced as a fine-grained data augmentation pipeline that linearly interpolates raw good speech pronunciations to generate mispronunciations at the phoneme level.

## 5 Evaluation Metrics

**Phoneme Error Rate (PER):** is a common metric used in the MD evaluation, measuring the accuracy of the predicted phoneme with the human-annotated sequence. However, PER might not provide a comprehensive assessment of model performance when mispronunciations are infrequent which is the case for MD datasets.

**Hierarchical Evaluation Structure:** The hierarchical evaluation structure developed in (Qian et al., 2010), has also been widely adopted in (Wang and Lee, 2015; Li et al., 2016a; EL Kheir et al., 2023a) among others. The hierarchical mispronunciation detection depends on detecting the misalignment over: *what is said* (annotated verbatim sequence); *what is predicted* (model output) along with *what should have been said* (text-dependent reference sequence). Based on the aforementioned sequences, the false rejection rate, false acceptance rate, and diagnostic error rate are calculated, using:

- True acceptance (**TA**): the number of phones annotated and recognized as correct pronunciations.
- True rejection (**TR**): the number of phones both annotated and correctly predicted as mispronunciations. The labels are further utilized to measure the diagnostic errors and correct diagnosis based on the prediction output and text-dependent canonical pronunciation.
- False rejection (**FR**): the number of phones wrongly predicted as mispronunciations.
- False acceptance (**FA**): the number of phones misclassified as correct pronunciations.

As a result, we can calculate the false rejection rate (**FRR**) that indicates the number of phones recognized as mispronunciations when the actual pronunciations are correct, false acceptance rate (**FAR**) that indicates phones misclassified as correct but are actually mispronounced, and diagnostic error rate (**DER**) using the following equations:

$$FRR = \frac{FR}{TA + FR} \quad (4)$$

$$FAR = \frac{FA}{FA + TR} \quad (5)$$

$$DER = \frac{DE}{CD + DE} \quad (6)$$

**Precision, Recall, and F-measure** are also widely used as the performance measures for mispronunciation detection. These metrics are defined as follows:

(2023) 他们引入了加权 DTW 对齐方法，作为 GOP 算法的替代方案，用于预测目标音素的概率及序列；其所提出的方法所得结果与 GOP 评分算法相当；同样，Anand等人(2023) 利用 DTW 分析了教师语音与学习者语音在 wav2vec 2.0 表征之间的对齐距离，以区分可懂语音与不可懂语音。

#### 4.8 数据增强

该领域面临的两大主要挑战是 L2 数据稀缺性以及负类（错误发音）的分布不均衡性。为应对这些挑战，研究人员采用了在语音评估领域已被证明相当有效的数据增强技术。此类方法采用了多种策略，例如通过引入不匹配的音素对来修改标准文本，同时保留原始词级语音数据（Fu 等，2021）。此外，还在特征空间中应用了混合技术，利用音素级的 GOP 池化方法构建词级训练数据（Fu等，2022）。另外，还利用聚类 SSL 模型嵌入向量的误差距离将音素声替换为相似的声音(Zhang 等，2022b)。后述这些方法依赖于对现有信息的复用，而非生成新的错误发音实例。在（Fernandez 等，2017）中，通过对音高、声道及声源特征进行语音变换来生成新样本。此外，L2-GEN可通过构建一种新型Seq2Seq音素 paraphrasing 模型（Zhang 等，2022a）来合成逼真的L2音素序列。Ko Rzekwa 等（2020b）提出了一种通过使用Neural TTS 生成错误重读单词的增强技术。此外，Korzekwa 等(2022)概述了采用三种方法生成错误发音的方案：音素-2-音素（P2P）方法依赖于对相应语音音频进行语音转写扰动；文本-2-语音方法生成与合成错误发音相匹配的语音信号；而语音-2-语音（S2S）方法则用于模拟语音韵律特征的不同方面。近期，SpeechBlender（EL Kheir 等，2023b）框架被引入作为一种细粒度数据增强流程，该流程通过线性插值原始正确语音发音来在音素层面生成错误发音。

## 5 评估指标

**音素错误率（PER）**：是机器翻译（MD）评估中常用的衡量指标，用于衡量预测音素与人工标注序列之间的匹配度。然而，当错误发音发生频率较低时（这在 MD 数据集场景下尤为常见），PER 可能无法全面反映模型的性能。

**分层评估结构**：由（Qian 等，2010）提出的分层评估结构，亦被广泛应用于（Wang 与Lee，2015；Li 等，2016a；EL Kheir 等，2023a）等研究中。分层错误发音检测依赖于对以下三者之间错位情况的检测：*实际说话语句*（标注的逐字序列）；*模型预测结果*；以及*应该说的话*（基于文本的参考序列）。基于上述序列，通过以下公式计算误拒率、误接受率及诊断错误率：

- **真实接受度（TA）**：被标注且被识别为正确发音的电话号码数量。
- **真实误判（TR）**：指同时被标注为误读且被正确预测为误读的电话号码。这些标签还可用于衡量诊断误差，并基于预测结果与文本依赖型标准发音对诊断结果进行校正。
- **误拒（FR）**：被错误预测为发音错误的电话号码。
- **错误接受（FA）**：被误判为正确发音的电话号码。

因此，我们可以利用以下公式计算误拒率（FRR）——该指标表示在实际发音正确的情况下仍被识别为错误发音的音素数量；误接受率（FAR）——该指标表示被误判为正确但实际为错误发音的音素数量；以及诊断错误率（DER）：

$$FRR = \frac{FR}{TA + FR} \quad (4)$$

$$FAR = \frac{FA}{FA + TR} \quad (5)$$

$$DER = \frac{DE}{CD + DE} \quad (6)$$

**精确率、召回率及 F 值**也广泛用作误读检测的性能评估指标。这些指标定义如下：

$$Precision = \frac{TR}{TR + FR} \quad (7)$$

$$Recall = \frac{TR}{TR + FA} = 1 - FAR \quad (8)$$

$$F - measure = 2 \cdot \frac{Precision \cdot Recall}{Precision + Recall} \quad (9)$$

**Pearson Correlation Coefficient:** PCC is widely used to measure the relation of the predicted score of fluency, stress, and prosody with other supra-segmental and pronunciation constructs with subjective human evaluation for pronunciation assessment. The human scores are typically averaged across all annotators to provide a comprehensive score.

## 6 Challenges and Future Look

There are two significant challenges facing advancing the research further: (1) the lack of public resources. Table 1 shows a handful of L2 languages. With 7,000 languages spoken on earth, there is an urgent need for inclusivity in pronouncing the languages of the world. (2) There is a need for a unified evaluation metric for pronunciation learning, this can be used to establish and continually maintain a detailed leaderboard system, which serves as a dynamic and multifaceted platform for tracking, ranking, and showcasing the advances in the field to guide researchers from academia and industry to push the boundaries for pronunciation assessments from unintelligible audio to native-like speech. The advent of AI technology represents a pivotal moment in our technological landscape, offering the prospect of far-reaching and transformative changes that have the potential to revolutionize a wide array of services in CAPT. Listing here some of the opportunities:

### Integration with Conversation AI Systems:

The progress made in Generative Pre-trained Transformer (GPT) led to a human-like text-based conversational AI. Furthermore, low-latency ASR has enhanced the adoption of speech processing in our daily life. Both have paved the way for the development of a reliable virtual tutor CAPT system, which is capable of interacting and providing students with instant and tailored feedback, thereby enhancing their pronunciation skills and augmenting private tutors.

**Multilingual:** Recent advancements in end-to-end ASR enabled the development of multi-lingual code-switching systems (Datta et al., 2020; Chowdhury et al., 2021; Ogunremi et al., 2023). The great progress in SSL expanded ASR capabilities

to support from over 100 (Pratap et al., 2023), to over 1,000 (Pratap et al., 2023) languages. Traditional research in pronunciation assessments focused on designing monolingual assessment systems. However, recent advancements in multilingualism allowed for the generalization of findings across different languages. Zhang et al. (2021a) explored the adaptation of pronunciation assessments from English (a stress-timed language) to Malay (a syllable-timed language). Meanwhile, Lin and Wang (2023b) investigated the use of language-specific embeddings for diverse languages, while optimizing the entire network within a unified framework.

**Children CAPT:** There is a noticeable imbalance in research between children learning pronunciation research papers, for example, reading assessments, compared to adults' L2 language learning. This disparity can be attributed to the scarcity of publicly available corpora and the difficulties in collecting children's speech data.

**Dialectal CAPT:** One implicit assumption in most of the current research in pronunciation assessment is that L2 is a language with a standard orthographic rule. However, Cases like dialectal Arabic – which is every Arab native language, there is no standard orthography. Since speaking as a native is the ultimate objective for advanced pronunciation learning, there is a growing demand for this task.

## 7 Conclusion

This paper serves as a comprehensive resource that summarizes the current research landscape in automatic pronunciation assessment covering both segmental and supra-segmental space. The paper offers insights into the following:

- Modeling techniques – highlighting design choices and their effect on performance.
- Data challenges and available resources – emphasizes the success of automatic data generation/augmentation pipeline and lack of consensus annotation guidelines and labels. The paper also lists available resources to the community along with the current state-of-the-art performances reported per resource.
- Importance of standardised evaluation metrics and steady benchmarking efforts.

With the current trend of end-to-end modeling and multilingualism, we believe this study will provide a guideline for new researchers and a foundation for future advancements in the field.

$$Precision = \frac{TR}{TR + FR} \quad (7)$$

$$Recall = \frac{TR}{TR + FA} = 1 - FAR \quad (8)$$

$$F - measure = 2 \cdot \frac{Precision \cdot Recall}{Precision + Recall} \quad (9)$$

**皮尔逊相关系数：**PCC 广泛用于衡量流利度、重音及韵律等预测得分与通过主观人工评估得出的其他超段位及发音相关构念之间的关系。人工评分通常通过对所有标注者的数据取平均值来得出综合评分。

## 6 挑战与未来展望

在进一步推进相关研究时，目前面临两大重大挑战：(1) 公共资源匮乏。表1列举了若干第二语言（L2）语言；鉴于全球共有 7,000 种在用语言，亟需在世界语言的发音标注与记录方面实现包容性。(2) 需要建立一套统一的发音学习评估指标体系；该体系可用于构建并持续维护一个详尽的排行榜系统，将其打造为一个动态且多维度的平台，用于追踪、排名及展示该领域的研究进展，从而引导学术界与产业界的研究人员突破发音评估的边界——从无法辨识的音频到接近母语水平的语音。人工智能技术的出现标志着我们技术发展版图中的一个关键转折点，它预示着一场深远且具有变革性的技术浪潮，有望彻底改变 CAPT 领域的众多服务形态。以下列举部分潜在机遇：

**与对话式人工智能系统的集成：**生成式预训练变换器（GPT）技术的突破催生了具备类人特性的基于文本的对话式人工智能系统；此外，低延迟语音识别技术的出现也促进了语音处理技术在日常生活中更广泛的应用。这两项技术为开发可靠的虚拟导师 CAPT 系统奠定了基础——该系统能够与学生进行实时交互并提供即时、个性化的反馈，从而提升学生的发音水平，并有效补充传统私人辅导服务。

**多语言支持：**端到端语音识别技术的最新进展推动了多语言代码切换系统的开发（Datta 等，2020；Chowdhury 等，2021；Ogunremi 等，2023）。自监督学习（SSL）领域的重大

进展扩展了语音识别能力，使其支持的语言数量从超过 100 种（Pratap 等，2023）扩展至超过 1,000 种（Pratap 等，2023）语言。传统的发音评估研究主要侧重于设计单语评估系统；然而，多语言处理领域的最新进展使得研究成果能够跨不同语言进行泛化。Zhang 等（2021a）探索了将英语（一种重音时序语言）的发音评估方法适配至马来语（一种音节时序语言）的应用。与此同时，Lin 与 Wang（2023b）研究了在统一框架下对整个网络进行优化的同时，针对多种语言采用语言特定嵌入表示的方法。

**儿童语言习得研究：**在相关研究中，针对儿童发音习得（例如阅读能力评估）的研究与针对成人第二语言习得的研究之间存在明显的不对称性。这种差异可归因于公开语料库的匮乏以及收集儿童语音数据所面临的困难。

**方言版 CAPT：**当前大多数语音产出评估研究中存在一个隐含假设，即第二语言（L2）都遵循标准的正字法规则。然而，实际情况中存在诸如方言阿拉伯语这类语言——所有阿拉伯语母语均无标准正字法体系。由于将口语表达作为母语水平的掌握是高级语音产出学习的最终目标，因此对这一学习任务的需求日益增长。

## 7 结论

本文是一份综合性资源，系统梳理了当前自动发音评估领域的研究现状，涵盖音段与超音段两个层面。本文对以下方面提供了深入见解：

- 建模技术——突出设计选择及其对性能的影响。
- 数据挑战与可用资源——着重阐述了自动数据生成/增强流水线的成功应用，以及当前在标注指南和标签标准方面尚缺乏统一共识的问题。本文还列出了面向学术界的各类可用资源，并介绍了针对每项资源所报告的当前最先进性能表现。
- 标准化评估指标与持续性基准测试工作的重要性。

顺应当前端到端建模与多语言化的发展趋势，我们相信本研究将为新晋研究人员提供指导，并为该领域的未来研究进展奠定基础。

## Limitations

In this overview, we address different constructs of pronunciation and various scientific approaches for detecting errors, predicting prosodic and fluency scores among others. However, we have not included the corrective feedback mechanism of CAPT system. Moreover, the paper does not cover, in detail, the limited literature on CAPT's user study, or other qualitative study involving subjective evaluation. With the fast-growing field of pronunciation assessments, it is hard to mention all the studies and resources. Therefore, we would also like to apologize for any oversights of corpora or major research papers in this study.

## Ethics Statement

We discussed publicly available research and datasets in our study. Any biases are unintended.

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## 局限性

在本综述中，我们探讨了发音相关的各类概念以及多种用于检测发音错误、预测韵律与流利度评分等的科学方法。然而，本文并未涵盖 CAPT 系统中的纠错反馈机制；此外，本文亦未对目前关于 CAPT 用户研究或其他涉及主观评估的定性研究这一有限的文献进行详细探讨。鉴于发音评估领域发展迅速，要全面列举所有相关研究与资源实属不易。因此，对于本研究中可能存在的语料库或重要学术论文遗漏之处，我们深表歉意。

## 伦理声明

我们在本研究中讨论了公开可获取的研究成果与数据集；任何偏倚均属非故意所致。

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## Appendix

### A Mispronunciation Assessment and Non-Native Datasets

In this section, we provide a comprehensive overview of existing pronunciation assessment datasets as presented in Table 1.

#### A.1 ISLE Speech Corpus (Menzel et al., 2000)

The ISLE corpus stands out as one of the largest speech corpora in terms of duration and offers the advantage of being distributed by ELDA. This corpus focuses on German and Italian accented English, featuring recordings of 23 intermediate-level speakers from each accent group. The participants, primarily employees and students from project sites in Italy, Germany, and the UK, were selected to achieve a balance of native languages (German/Italian) while including a small number of non-native speakers from other countries (Spanish, French, Chinese) and native British English speakers for comparison purposes. The corpus contains readings of both nonfictional autobiographic text (1300 words) and short utterances (1100 words) designed to cover common pronunciation errors made by language learners. It offers annotations at both the word and phone levels, making it particularly valuable for developing Computer Assisted Language Learning systems. The annotation process involved multiple steps, including quality checks, reference transcription, forced alignment, and the addition of canonical pronunciations and stress markings. An emphasis was placed on matching non-English phones to the closest equivalent in the UK English phone set, with occasional input from a trained phonetician and a native speaker of the speaker's mother tongue for verification and quality improvement purposes.

#### A.2 English read by Japanese Corpus (ERJ) (Minematsu et al., 2004)

The ERJ corpus (English Read by Japanese) is a database of English speech read by Japanese students. It was created to support research in computer-assisted language learning (CALL). The corpus contains 800 utterances from 202 (100 males and 102 females) Japanese university students, each of whom read a set of 100 sentences. The sentences were selected to be phonetically balanced and to cover a variety of grammatical structures. The corpus is annotated with phonemic transcriptions and prosodic markings. The ERJ corpus

consists of two sets of data: a phonemic pronunciation set and a prosody set. The phonemic pronunciation set contains 460 phonetically-balanced sentences, 32 sentences including phoneme sequences difficult for Japanese to pronounce correctly, and 100 sentences designed for the test set. The prosody set contains 94 sentences with various intonation patterns, 120 sentences with various accent and rhythm patterns, and 109 words with various accent patterns. The ERJ corpus is annotated with phonemic transcriptions and prosodic markings. The phonemic transcriptions are based on the International Phonetic Alphabet (IPA). The prosodic markings include information about intonation, accent, and rhythm.

#### A.3 Chinese University Chinese Learners of English (CU-CHLOE) (Meng et al., 2007a)

The CU-CHLOE corpus encompasses a diverse group of speakers, including 110 Mandarin speakers (60 males and 50 females) and 100 Cantonese speakers (50 males and 50 females). It is structured into five distinct sections, namely confusable words, minimal pairs, phonemic sentences, the Aesop's Fable "The North Wind and the Sun," and prompts sourced from the TIMIT dataset. Trained linguists have diligently labeled all sections, except for the TIMIT prompts, contributing to approximately 30% of the comprehensive CHLOE data. This expert annotation ensures the corpus provides reliable and precise linguistic information, making it a valuable resource for exploring Mandarin and Cantonese speech characteristics and facilitating advancements in research within these languages.

#### A.4 EURONOUNCE (Cylwik et al., 2009)

EURONOUNCE is a speech corpus developed for an ASR-based pronunciation tutoring system. The annotation conveys a phonetic segmentation, with identifying pronunciation errors, including substitutions, insertions, and deletions. The resulting annotations are then reviewed by a native speaker of the source language (German) to validate the assessment.

#### A.5 iCALL (Chen et al., 2015)

iCALL is comprised of 90,841 spoken statements delivered by 305 individuals, spanning a cumulative duration of 142 hours. The speaker composition ensures gender equality, encompasses various native languages, and reflects a representative range of adult Mandarin learners. These oral statements

## 附录

### A 发音错误评估与非母语者语料库

本节对现有的发音评估数据集进行了全面概述，具体如表1所示。

#### A.1 ISLE语音语料库 (Menzel 等, 2000)

ISLE语料库是规模最大的语音语料库之一（按时长计），其优势在于由 ELDA 公司负责分发。该语料库专注于带有德语和意大利语口音的英语语音数据，收录了来自每个口音组的23名中等水平发音者的录音样本。参与人员主要来自意大利、德国和英国各项目实施地的员工及学生；选择这些参与者旨在实现母语（德语/意大利语）使用者与少量来自其他国家（西班牙语、法语、中文）的非母语使用者以及母语为英式英语的使用者之间的均衡分布，以便进行对比研究。该语料库包含非虚构自传体文本（1300词）和短句（1100词）的朗读样本，旨在涵盖语言学习者常见的发音错误。该语料库提供了词级和音素级的标注信息，因此对于开发计算机辅助语言学习（CFL）系统具有重要价值。标注过程包含多个步骤，包括质量检查、参考转写、强制对齐以及添加规范发音和重音标记。重点在于将非英语语音样本与英式英语语音集中的最接近对应项进行匹配；为确保准确性及提升质量，该过程偶尔会邀请经过培训的语音学专家及该说话者母语母语者参与验证与质量优化工作。

#### A.2 日语语料库朗读的英文文本 (ERJ) (Minematsu 等, 2004)

ERJ 语料库（《日语朗读英语》）是一个由日本学生朗读的英语语音数据库，旨在为计算机辅助语言学习（CALL）领域的研究提供支持。该语料库收录了来自202名日本大学生（其中男性100名，女性102名）的800条语音片段，每位参与者均朗读了100条句子。这些句子经过精心筛选，确保语音平衡且涵盖多种语法结构。该语料库均配有音素转写标注和韵

律标记。ERJ 语料库由两组数据构成：音素发音集和韵律集。音素发音集包含460条语音平衡的句子、32条包含日本语学习者难以正确发音的音素序列的句子，以及100条专为测试集设计的句子；韵律集则包含94条具有不同语调模式的句子、120条具有不同重音与节奏模式的句子，以及109个具有不同重音模式的单词。该语料库均配有音素转写标注和韵律标记，其中音素转写基于国际音标（IPA）。韵律标记包含关于语调、重音和节奏的信息。

#### A.3 中国高校英语学习者项目 (CU-Chloe) (Meng 等, 2007a)

CU-Chloe语料库涵盖了一组多样化的说话者群体，包括110名普通话使用者（60名男性和50名女性）以及100名粤语使用者（50名男性和50名女性）。该语料库分为五个独立部分：易混淆词汇、最小对立对、音位句、伊索寓言《北风与太阳》以及源自 TIMIT 数据集的提示语。经过专业训练的语言学家对所有部分（除 TIMIT 提示语外）均进行了细致标注，这些标注内容约占整个Chloe语料库的30%。这一专业标注工作确保了该语料库所提供的语言学信息可靠且精确，使其成为探究普通话与粤语语音特征、推动相关语言学研究发展的宝贵资源。

#### A.4 euronounce(Cylwik 等, 2009)

EURONOUNCE 是一个专为基于自动语音识别（ASR）的发音辅导系统开发的语音语料库。其标注内容包含音系学分段信息，并对发音错误（包括音素替换、插入和删除）进行标识。随后，这些标注内容将由母语为德语的母语者进行审核，以验证其准确性。

#### A.5 iCALL (Chen 等, 2015)

iCALL 包含由 305 名参与者提供的 90,841 条口语陈述，累计时长为 142 小时。参与者的构成实现了性别均衡，涵盖多种母语，并反映了成年 Mandarin 学习者的代表性群体。这些口语陈述已进行音标转写，并由熟练的 Mandarin

have been transcribed phonetically and evaluated for fluency by proficient native Mandarin speakers.

#### **A.6 SingaKids-Mandarin (Chen et al., 2016b)**

SingaKids-Mandarin is a comprehensive speech corpus consisting of recordings from 255 Singaporean children between the ages of 7 and 12. The corpus aims to provide a resource for studying Mandarin Chinese pronunciation and language acquisition in Singaporean children. The corpus contains a total of 125 hours of audio data, with 75 hours dedicated to speech. Within this dataset, there are 79,843 utterances, each of which has been meticulously annotated by human experts. The annotations include phonetic transcriptions, lexical tone markings, and proficiency scoring at the level of individual utterances. The reading scripts used in the corpus encompass a wide range of utterance styles. They cover syllable-level minimal pairs, individual words, phrases, complete sentences, and even short stories. This diversity allows for a thorough analysis of different aspects of Mandarin pronunciation and fluency in Singaporean children.

#### **A.7 SHEFCE (Ng et al., 2017b)**

SHEFCE (ShefCE) is a bilingual parallel speech corpus that focuses on Cantonese and English. It was recorded by second language (L2) English learners in Hong Kong. The corpus consists of recordings from 31 undergraduate to postgraduate students, aged 20 to 30. The corpus includes a total of 25 hours of speech data, with approximately 12 hours recorded in Cantonese and 13 hours recorded in English. The primary goal of this corpus is to provide a resource for studying the speech patterns, pronunciation, and language acquisition of Cantonese-speaking individuals who are learning English as a second language.

#### **A.8 L2-ARCTIC (Zhao et al., 2018a)**

The L2-ARCTIC<sup>2</sup> corpus is a specialized speech corpus designed for research in voice conversion, accent conversion, and mispronunciation detection in non-native English. It encompasses a substantial collection of 26867 utterances from 24 non-native speakers (12 males and 12 females) whose *L1* languages include Hindi, Korean, Mandarin, Spanish, Arabic, and Vietnamese. The recordings were sourced from a total of 4 speakers per *L1* language, consisting of 2 males and 2 females ensuring a balanced distribution in terms of gender and native

languages (*L1*s). Yet, only 150 utterances is manually per speaker to identify three types of segmental mispronunciation errors: substitutions, deletions, and insertions resulting in 3.66 hours.

#### **A.9 VoisTUTOR corpus (Yarra et al., 2019)**

VoisTUTOR is a pronunciation assessment corpus of Indian second language (L2) learners learning English. The corpus consists of audio recordings of 16 Indian L2 learners reading a set of 1676 sentences. The recordings are accompanied by phonetic transcriptions, human ratings of pronunciation accuracy on a scale of 0 to 10 for each utterance, and binary decisions for seven factors that affect pronunciation quality: intelligibility, phoneme quality, phoneme mispronunciation, syllable stress quality, intonation quality, correctness of pauses, and mother tongue influence.

#### **A.10 SELL-CORPUS (Chen et al., 2019)**

SELL-CORPUS is a multiple accented speech corpus for L2 English learning in China. The corpus consists of audio recordings of 389 volunteer speakers, including 186 males and 203 females. The speakers are from seven major regional dialects of China, including Mandarin, Cantonese, Wu, Min, Hakka, and Southwestern Mandarin. The corpus contains 31.6 hours of speech recordings. Each recording in the corpus contains a word-level orthographic transcription manually inspected and cleaned by inserting, substituting, or deleting mismatching characters.

#### **A.11 English Pronunciation by Argentinians Database (EpaDB) (Vidal et al., 2019a)**

EpaDB consists of English phrases recorded by native Spanish speakers with varying levels of English proficiency. The recordings are annotated with ratings indicating the quality of pronunciation at the phrase level. Additionally, detailed phonetic alignments and transcriptions are provided, indicating which phones were actually pronounced by the speakers.

#### **A.12 Speechocean762 (Zhang et al., 2021b)**

Speechocean762 is an extensive dataset specifically designed for pronunciation assessment. It comprises a total of 5,000 English utterances obtained from 250 non-native speakers. Each utterance in the dataset is associated with five aspect scores at the utterance level, namely accuracy, fluency, completeness, prosody, and a total score ranging from 0 to 10. Additionally, for each word within the ut-

<sup>2</sup>version 5 released in 2020 available: <https://psi.engr.tamu.edu/l2-arctic-corpus>

母语者对其流畅度进行了评估。

#### A.6 SingaKids-Mandarin (Chen 等, 2016b)

SingaKids-Mandarin 是一个综合性汉语语音语料库, 收录了 255 名年龄在 7 至 12 岁之间的新加坡儿童的语音录音。该语料库旨在为研究新加坡儿童的普通话发音及语言习得过程提供一个研究资源。该语料库共包含 125 小时的音频数据, 其中 75 小时为语音样本; 在该数据集中共有 79,843 条话语片段, 每条话语片段均经过人类专家的细致标注。标注内容包括音标转写、词汇声调标记以及针对单条话语片段的熟练度评分。该语料库所采用的朗读文本涵盖了广泛的语句风格, 包括音节层面的最小对立对、单个单词、短语、完整句子乃至短篇故事。这种多样性使得对新加坡儿童普通话发音及流利度的各个方面进行深入分析成为可能。

#### A.7 SHEFCE(Ng 等, 2017b)

SHEFCE (ShefCE) 是一个双语平行语音语料库, 专注于粤语与英语。该语料库由香港的英语二语 (L2) 学习者录制, 收录了 31 名年龄介于 20 至 30 岁之间的本科生及研究生的语音样本。该语料库共包含 25 小时的语音数据, 其中约 12 小时为粤语录音, 13 小时为英语录音。本语料库的主要目的是为研究以粤语为母语、将英语作为第二语言学习者的语音模式、发音及语言习得过程提供一个研究资源。

#### A.8 L2-北极地区 (Zhao 等, 2018a)

L2-arctic<sup>2</sup>语料库是一个专为语音转换、口音转换以及非母语英语中的发音错误检测研究而设计的专用语音语料库。该语料库包含由 24 名非母语英语使用者 (12 名男性和 12 名女性) 提供的 26,867 条语音样本, 这些使用者的 L1 语言涵盖印地语、韩语、普通话、西班牙语、阿拉伯语和越南语。这些录音采集自每种 L1 语言对应的 4 名说话者, 其中包含 2 名男性和 2 名女性, 从而确保了在性别和母语背景方面具有均衡的分布。

<sup>2</sup>第 5 版于 2020 年发布, 下载地址: <https://psi.engr.tamu.edu/l2-arctic-corpus>

语言 (L1s)。然而, 仅对每位说话者手动标注了 150 条话语, 用于识别三种类型的音段性发音错误——替换、删除和插入——共计 3.66 小时。

#### A.9 VoisTUTOR语料库 (Yarra 等, 2019)

VoisTUTOR 是一个针对印度英语第二语言 (L2) 学习者的发音评估语料库。该语料库包含 16 名印度 L2 学习者朗读 1676 个句子的音频录音; 每段录音均附有语音转写文本、针对每个语句采用 0 至 10 分制进行的人工发音准确度评分, 以及针对影响发音质量的七个因素所作出的二元判定: 语义清晰度、音素质量、音素误读、音节重音质量、语调质量、停顿正确性以及母语影响。

#### A.10 SELL-corpus (Chen 等, 2019)

SELL-corpus 是一个专为中国英语作为第二语言 (L2) 学习设计的、包含多重重音语音语料库。该语料库收录了 389 名志愿发音者的语音录音, 其中包括 186 名男性和 203 名女性; 这些发音者来自中国七大主要地方方言, 涵盖普通话、粤语、吴语、闽语、客家语及西南官话。该语料库共包含 31.6 小时的语音录音。语料库中的每段录音均包含一份词级正字法转写文本, 该转写文本经过人工校对与清理, 过程中通过插入、替换或删除不匹配的汉字来完成校正。

#### A.11 阿根廷人英语发音数据库 (EpaDB) (Vidal 等, 2019a)

EpaDB 包含由英语水平各异的西班牙语母语者录制的英语短语。这些录音均附有标注, 通过评分方式反映各短语的发音质量; 此外, 还提供了详细的音系对齐与转写结果, 用于标明说话者实际发出的音素。

#### A.12 Speechocean762(Zhang 等, 2021b)

Speechocean762 是一个专为发音评估设计的大型语料库。该语料库共包含 5,000 条英语话语样本, 这些样本采集自 250 名非母语者。语料库中的每条话语在话语层面对应五个方面评分——即准确度、流利度、完整性、韵律性以及一个范围在 0 到 10 之间的总分; 此外, 针对话语中的每个单词还提供三个方面评分——包

terance, three aspect scores are provided, including accuracy, stress, and a total score ranging from 0 to 10. Furthermore, an accuracy score is assigned to each individual phoneme, ranging from 0 to 2. To ensure reliability, each of these scores is annotated by five expert evaluators.

### A.13 LATIC (ZHANG, 2021)

LATIC primarily targets non-native learners of Mandarin Chinese. The dataset comprises four hours of recordings involving specifically selected non-native Chinese speakers. The participants' L1's vary, including Russian, Korean, French, and Arabic. Following each audio file, annotators transcribed the "closest" transcript and provided modern Mandarin annotations after careful listening.

### A.14 Arabic-CAPT (Algabri et al., 2022)

Arabic-CAPT is an Arabic mispronunciation detection corpus consisting of 62 non-native Arabic speakers from 20 different nationalities, totaling 2.36 hours of speech data. The Arabic non-native speech is annotated following the guidelines in (Zhao et al., 2018a).

### A.15 AraVoiceL2 (EL Kheir et al., 2023b)

AraVoiceL2 is an Arabic mispronunciation detection corpus comprised of 5.5 hours of data recorded by 11 non-native Arabic speakers. Each speaker recorded a fixed list of 642 words and short sentences, making for a total of 7,062 recordings. The corpus is annotated at character level including diacritics following (Zhang et al., 2021b) guidelines.

### A.16 Non-Native Datasets:

Table 2 provides a comprehensive overview of existing non-native datasets that are particularly beneficial as they enable the extraction of error patterns allowing for a thorough assessment of L2 pronunciation. These datasets can also be used to train robust ASR models, from which we can extract valuation features to accurately score L2 speech. Furthermore, non-native datasets can enhance existing pronunciation assessment end-to-end approaches.

## B Annotation

In this section, we provide an overview of the standard approaches to annotate segmental and supra-segmental errors widely used in MDD research.

### B.1 Segmental Annotation

Segmental human annotation can be approached from two perspectives. The first and the commonly utilized approach in most available MDD

corpora involves linguistics experts transcribing the actual sequence of phonemes spoken by the learner (Bonaventura et al., 2000; Zhao et al., 2018b; Vidal et al., 2019b). The resulted transcription is commonly referred as hypothesis annotation. Additionally, extra tasks can be incorporated, such as providing time boundaries for each pronounced phoneme, to further enhance the annotation process. This approaches may have limitations in capturing non-clear speech instances, such as heavily accented pronunciations that may not be easily detected by human annotators. This leads to the second approach, which incorporates scoring-based methods in addition to hypothesis annotation (Zhang et al., 2021c). In this approach, a score is assigned to each phoneme: 0 represents deleted or mispronounced phonemes, 1 indicates heavily accented pronunciation, and 2 signifies good pronunciation. This scoring-based approach provides a more comprehensive assessment of pronunciation quality, particularly in cases where clear detection by human annotators may be challenging.

### B.2 Supra-segmental Annotation

Limited research has been conducted regarding the annotation of supra-segmental features at the rhythm, stress, and intonation levels such as in (Arvaniti and Baltazani, 2000; Chen et al., 2016b; Cole et al., 2017). However, the ultimate objective of annotating these supra-segmental aspects is to ensure the fluency and intelligibility of L2 learners' speech. Hence the most commonly used annotated datasets at the prosodic level provide human-scored words, and sentences based on overall pronunciation quality and fluency. Multiple tiers of human scoring annotations can be applied in this context. This includes providing the accuracy of pronounced words to assess their intelligibility, assigning scores to evaluate the positioning of stress within individual words or within sentences, and evaluating sentence fluency by considering factors such as pauses, repetitions, and stammering in speech as adapted in (Zhang et al., 2021c).

括准确度、重音度以及一个范围在 0 到 10 之间的总分。同时，每个独立音素均被赋予一个范围在 0 到 2 之间的准确度评分。为确保评分结果的可靠性，上述各项评分均由五位专家评估员进行标注。

### A.13 LATIC (张, 2021)

LATIC 主要面向非母语中文学习者。该数据集包含时长为 4 小时的录音资料，这些录音由经过专门筛选的非母语中文母语者录制。参与者的母语背景多样，包括俄语、韩语、法语和阿拉伯语。在每段音频文件结束后，标注员会仔细聆听，并转录“最接近”的文本，并为其添加现代标准汉语标注。

### A.14 Arabic-CAPT(Algabri 等, 2022)

Arabic-CAPT 是一个阿拉伯语发音错误检测语料库，包含来自 20 个不同国籍的 62 名非阿拉伯语母语者，总语音数据时长为 2.36 小时。该非阿拉伯语母语语音数据的标注遵循了(Zhao 等, 2018a)中提出的指导原则。

### A.15 AraVoiceL2 (EL Kheir 等, 2023b)

AraVoiceL2 是一个阿拉伯语发音错误检测语料库，包含由 11 名非阿拉伯语母语者录制的 5.5 小时数据。每位说话者均录制了包含 642 个单词及短句的固定词表，共计 7,062 段录音。该语料库在字符级别进行了标注，包括根据(Zhang 等, 2021b)指南所规定的变音符号。

### A.16 非原生数据集:

表2全面概述了现有的非母语语料库；这些语料库尤为实用，因为它们支持提取错误模式，从而能够对第二语言(L2)发音进行深入评估。这些语料库还可用于训练鲁棒的自动语音识别(ASR)模型，进而从中提取估值特征以对L2语音进行精准评分。此外，非母语语料库还能增强现有的端到端发音评估方法。

## B 注释

本节概述了在 MDD 研究中广泛采用的对段落级及超段落级错误进行标注的标准方法。

### B.1 分段标注

段落式人类标注可从两个视角进行探讨。第一种方法在大多数现有的 MDD 语料库中被广泛采用，即由语言学专家对学习者的实际发出的音

素序列进行转写(Bonaventura 等, 2000; Zhao 等, 2018b; Vi dal 等, 2019b)。由此产生的转写结果通常被称为“假设标注”。此外，还可引入额外任务，例如为每个发音音素标注时间边界，以进一步优化标注过程。然而，该方法在捕捉非清晰语音实例(如带有强烈重音的发音，这类发音可能不易被人工标注员识别)时可能存在局限性。这引出了第二种方法，即在假设标注基础上结合基于评分的方法(Zhang 等, 2021c)。在该方法中，为每个音素分配一个评分：0代表被删除或发音错误的音素，1表示带有强烈重音的发音，2则表示发音良好。这种基于评分的方法能够对发音质量进行更全面的评估，尤其适用于人工标注人员难以清晰识别的场景。

### B.2 超节段标注

目前针对节奏、重音及语调层面的超段位特征标注的研究尚不充分，例如在(Ar vaniti 与 Baltazani, 2000; Chen 等, 2016b; Cole 等, 2017)中。然而，标注这些超段位特征的最终目的是确保二语学习者言语的流利度与可懂度。因此，最常用的韵律层面标注数据集通常提供人工评分的单词及句子，这些评分基于整体发音质量与流利度。在此背景下，可采用多层次的人工评分标注方法，包括：通过提供发音单词的准确度来评估其可懂度；为评估单个单词或句子内重音位置而赋予评分；以及通过考虑言语中的停顿、重复和口吃等因素来评估句子流利度(如(Zhang 等, 2021c)所采用的方法)。

| Corpus                                                                               | Languages          | Dur / #Utt     | #Speakers |
|--------------------------------------------------------------------------------------|--------------------|----------------|-----------|
| Demuth Sesotho Corpus ( <a href="#">Demuth, 1992</a> )                               | Sesotho            | 98h / / 13250  | 4         |
| TIDIGITS ( <a href="#">Leonard and Doddington, 1993</a> )                            | English            | -              | / 326     |
| CMU Kids Corpus ( <a href="#">Eskenazi et al., 1997</a> )                            | English            | / 5180         | 76        |
| CU Children’s Read and Prompted Speech Corpus ( <a href="#">Hagen et al., 2003</a> ) | English            | / 100          | 663       |
| CU Story Corpus ( <a href="#">Hagen et al., 2003</a> )                               | English            | 40h / 7062     | 106       |
| PF-STAR Children’s Speech Corpus ( <a href="#">Batliner et al., 2005</a> )           | English            | 14.5h /        | 158       |
| TBALL ( <a href="#">Kazemzadeh et al., 2005</a> )                                    | English            | 40h / 5000     | 256       |
| Swedish NICE Corpus ( <a href="#">Bell et al., 2005</a> )                            | Swedish            | -              | 5580      |
| Providence Corpus ( <a href="#">Demuth et al., 2006</a> )                            | English            | 363h /         | 6         |
| Lyon Corpus ( <a href="#">Demuth and Tremblay, 2007</a> )                            | French             | 185h /         | 4         |
| CHIEDE ( <a href="#">Garrote, 2008</a> )                                             | Spanish            | 8h / / 15444   | 59        |
| CFSC ( <a href="#">Pascual and Guevara, 2012</a> )                                   | Filipino           | 8h /           | 57        |
| CASSCHILD ( <a href="#">Gao et al., 2012</a> )                                       | Mandarin           | -              | 23        |
| CALL-SLT ( <a href="#">Rayner et al., 2014</a> )                                     | German             | / 5000         | -         |
| Boulder Learning—MyST Corpus ( <a href="#">Boulder Learning Inc, 2019</a> )          | English            | 393h / 228874  | 1371      |
| TLT-school ( <a href="#">Gretter et al., 2020</a> )                                  | English and German | 119.1h / 26059 | 6547      |

Table 2: Non-Native Speech Datasets

| 语料库                                                    | 语言    | Dur / #Utt     | #演讲者  |
|--------------------------------------------------------|-------|----------------|-------|
| Demuth Sesotho 语料库 (Demuth, 1992)                      | 塞索托语  | 98h / / 13250  | 4     |
| Tidigits(Leonard 与 Doddington, 1993)                   | 英语    | -              | / 326 |
| CMU 儿童语料库 (Eskenazi 等, 1997)                           | 英语    | / 5180         | 76    |
| CU儿童阅读与引导性言语语料库(Hagen 等, 2003)                         | 英语    | / 100          | 663   |
| CU故事语料库(Hagen 等, 2003)                                 | 英语    | 40h / 7062     | 106   |
| PF-STAR 儿童言语语料库 (Batliner 等, 2005)                     | 英语    | 14.5h /        | 158   |
| TBALL (Kazemzadeh 等, 2005)                             | 英语    | 40h / 5000     | 256   |
| 瑞典 NICE 语料库 (Bell 等, 2005)                             | 瑞典的   | -              | 5580  |
| Providence Corpus (Demuth 等, 2006)                     | 英语    | 363h /         | 6     |
| Lyon Corpus(Demuth 与 Tremblay, 2007)                   | 法国的   | 185h /         | 4     |
| CHIEDE (Garrote, 2008)                                 | 西班牙的  | 8h / / 15444   | 59    |
| CFSC (Pascual 与 Guevara, 2012)                         | 菲律宾人  | 8h /           | 57    |
| Casschild (Gao 等, 2012)                                | 普通话   | -              | 23    |
| CALL-SLT(Rayner 等, 2014)                               | 德国的   | / 5000         | -     |
| Boulder Learning—MyST 语料库 (Boulder Learning Inc, 2019) | 英语    | 393h / 228874  | 1371  |
| TLT -School (Gretter 等, 2020)                          | 英语与德语 | 119.1h / 26059 | 6547  |

表2: 非母语语音数据集